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Recommendation

ITU-T G.8275.1 (02/2026)

SERIES G: Transmission systems and media, digital systems and networks

Packet over Transport aspects – Synchronization, quality and availability targets

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Precision time protocol telecom profile for phase/time synchronization with full timing support from the network

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Recommendation ITU-T G.8275.1

Precision time protocol telecom profile for phase/time synchronization with full timing support from the network

Summary

Recommendation ITU-T G.8275.1 specifies a profile for telecommunication applications based on the precision time protocol (PTP) as defined in IEEE 1588. The profile specifies the PTP functions that are necessary to ensure network element interoperability for the delivery of accurate phase/time (and frequency) synchronization. The profile is based on the use of full timing support (FTS) from the network architecture described in Recommendation ITU-T G.8274 and definitions described in Recommendation ITU-T G.8260. Common aspects of PTP profiles for phase/time synchronization are specified in Recommendation ITU-T G.8275.

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Recommendation ITU-T G.8275.1

Precision time protocol telecom profile for phase/time synchronization with full timing support from the network

1 Scope

This Recommendation specifies a profile for telecommunication applications based on the precision time protocol (PTP) as defined in [IEEE 1588]. The profile specifies the PTP functions that are necessary to ensure network element interoperability for the delivery of accurate phase/time (and frequency) synchronization. The profile is based on the full timing support from the network architecture described in [ITU-T G.8274] and definitions described in [ITU-T G.8260]. Common aspects of PTP profiles for phase/time synchronization are specified in [ITU-T G.8275].

This version of the profile specifies the high-level design requirements, modes of operation for the exchange of PTP messages, the PTP protocol mapping, the best timeTransmitter clock algorithm (BTCA) options, as well as the PTP protocol configuration parameters.

NOTE – The parameters defined in this version of the profile are chosen based on the case where physical layer frequency support is provided, and the case without physical layer frequency support (i.e., PTP only) is for further study.

This Recommendation also specifies some aspects necessary for use in a telecom environment that are outside the scope of, and complement, the PTP profile.

An implementation compliant with this profile can claim compliance with either IEEE Std 1588-2008 [IEEE 1588-2008] or IEEE Std 1588-2019 [IEEE 1588-2019]. Considerations on the use of one or the other profile are provided in Appendix XIII of this Recommendation. Compliance for a specific implementation with either of the versions [IEEE 1588-2008] or [IEEE 1588-2019] of the standard should be stated when referring to this profile.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T G.709] Recommendation ITU-T G.709/Y.1331 (2020), *Interfaces for the optical transport network.*

[ITU-T G.709.1] Recommendation ITU-T G.709.1 (2024), *Flexible OTN common elements.*

[ITU-T G.781] Recommendation ITU-T G.781 (2026), *Synchronization layer functions for frequency synchronization based on the physical layer.*

[ITU-T G.810] Recommendation ITU-T G.810 (1996), *Definitions and terminology for synchronization networks.*

[ITU-T G.7041] Recommendation ITU-T G.7041/Y.1303 (2016), *Generic framing procedure.*

[ITU-T G.8260] Recommendation ITU-T G.8260 (2022), *Definitions and terminology for synchronization in packet networks.*

[ITU-T G.8265.1] Recommendation ITU-T G.8265.1/Y.1365.1 (2022), *Precision time protocol telecom profile for frequency synchronization.*

[ITU-T G.8271.1] Recommendation ITU-T G.8271.1/Y.1366.1 (2022), *Network limits for time synchronization in packet networks with full timing support from the network.*

[ITU-T G.8272] Recommendation ITU-T G.8272 (2025), *Timing characteristics of primary reference time clocks.*

[ITU-T G.8272.1] Recommendation ITU-T G.8272.1 (2024), *Timing characteristics of enhanced primary reference time clocks.*

[ITU-T G.8272.2] Recommendation ITU-T G.8272.2 (2024), *Timing characteristics of coherent network primary reference time clocks.*

[ITU-T G.8273] Recommendation ITU-T G.8273/Y.1368 (2023), *Framework of phase and time clocks.*

[ITU-T G.8273.2] Recommendation ITU-T G.8273.2/Y.1368.2 (2023), *Timing characteristics of telecom boundary clocks and telecom time synchronous clocks for use with full timing support from the network.*

[ITU-T G.8273.3] Recommendation ITU-T G.8273.3/Y.1368.3 (2020), *Timing characteristics of telecom transparent clocks for use with full timing support from the network.*

[ITU-T G.8274] Recommendation ITU-T G.8274 (2025), *Architecture and requirements for packet-based time and phase distribution.*

[ITU-T G.8275] Recommendation ITU-T G.8275 (2026), *Common aspects of PTP profiles for phase/time synchronization.*

[ITU-T G.8312] Recommendation ITU-T G.8312 (2020), *Interfaces for metro transport networks.*

[IEEE 802.1Q] IEEE 802.1Q-2022, *IEEE Standard for Local and Metropolitan Area Networks – Media Access Control (MAC) Bridges and Virtual Bridged Local Area Networks*.

[IEEE 1588] Either [IEEE 1588-2008] or [IEEE 1588-2019] depending on the specific implementation. See clause 5 of Conventions for more details.

[IEEE 1588-2008] IEEE 1588-2008, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*.

[IEEE 1588-2019] IEEE 1588-2019, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*.

[IEEE 1588a-2023] IEEE 1588a-2023 Amendment 3, *Precision Time Protocol (PTP) Enhancements for Best Master Clock Algorithm (BMCA) Mechanisms.*

[OIF FLEXE IA] Optical Internetworking Forum, IA OIF-FLEXE-03.0 (2025), *FlexE 3.0 Implementation Agreement*.

3 Definitions

3.1 Terms defined elsewhere

The terms and definitions used in this Recommendation are contained in [ITU-T G.810] and [ITU-T G.8260].

3.2 Terms defined in this Recommendation

This Recommendation defines the following term:

3.2.1 special precision time protocol port: A precision time protocol (PTP) port that translates between the native timing mechanism of a medium and PTP.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

1 PPS	One Pulse Per Second
AS	Available Seconds
BC	Boundary Clock
BTCA	Best TimeTransmitter Clock Algorithm
cnPRTC	coherent network Primary Reference Time Clock
cTE	constant Time Error
Cth	Clockclass threshold
D1S	Degraded 1 Second
DS	Dataset
DSLAM	Digital Subscriber Line Access Multiplexer
dTE	dynamic Time Error
ePRC	enhanced Primary Reference Clock
ePRTC	enhanced Primary Reference Time Clock
EUI	Extended Unique Identifier
FlexE	Flex Ethernet
FlexO	Flex OTN
FTS	Full Timing Support
GFP-F	Generic Framing Procedure-Framed
GM	Grandmaster
GNSS	Global Navigation Satellite System
HRM	Hypothetical Reference Model
ID	Identifier
IWF	Interworking Function
LAG	Link Aggregation
LTR	Local Time Reference
MAC	Media Access Control
MA-L	MAC Address – Large
MA-M	MAC Address – Medium
MA-S	MAC Address – Small
maxTE	maximum Time Error

MTN	Metro Transport Network
N1S	Nominal 1 Second
NMS	Network Management System
OC	Ordinary Clock
OLT	Optical Line Terminal
ONU	Optical Network Unit
OSC	Optical Supervisory Channel
OSMC	OTN Synchronization Messaging Channel
OTN	Optical Transport Network
OTUk	Optical Transport Unit, level k
OUI	Organizationally Unique Identifier
PRC	Primary Reference Clock
PRS	Primary Reference Source
PRTC	Primary Reference Time Clock
PTP	Precision Time Protocol
PTSF	Packet Timing Signal Fail
PTSF-LOTM	Packet Timing Signal Fail – Loss Of Timing Messages
PTSF-SU	Packet Timing Signal Fail – Synchronization Uncertain
QL	Quality Level
RT-DSLAM	Remote Terminal Digital Subscriber Line Access Multiplexer
SDO	Standards Development Organization
SEC	Synchronous Equipment Clock
SF	Signal Fail
SSU	Synchronization Supply Unit
SSU-A	primary level SSU
SSU-B	secondary level SSU
ST2	Stratum 2
ST3	Stratum 3
ST3E	Stratum 3 Enhanced
SyncE	Synchronous Ethernet
T-BC	Telecom Boundary Clock
TC	Transparent Clock
TDEV	Time Deviation
TE	Time Error
T-GM	Telecom Grandmaster
TLV	Type Length Value
ToD	Time of Day

T-TC	Telecom Transparent Clock
T-TSC	Telecom Time Synchronous Clock
UAS	Unavailable Seconds
UTC	Coordinated Universal Time
VLAN	Virtual Local Area Network
WTR	Wait To Restore
xDSL	x-type Digital Subscriber Line
xPON	x-type Passive Optical Network

5 Conventions

Within this Recommendation, the following conventions are used: the term precision time protocol (PTP) refers to the PTP defined in [IEEE 1588]. PTP messages used within this Recommendation are defined in [IEEE 1588] and are identified using italicized text.

The term telecom boundary clock (T-BC) refers to a device consisting of a boundary clock (BC) as defined in [IEEE 1588] and this Recommendation, with additional performance characteristics defined in [ITU-T G.8273.2].

The term telecom transparent clock (T-TC) refers to a device consisting of a transparent clock (TC) as defined in [IEEE 1588] and this Recommendation, with additional performance characteristics defined in [ITU-T G.8273.3].

The term telecom grandmaster (T-GM) refers to a device consisting of a grandmaster (GM) clock as defined in [IEEE 1588] and this Recommendation, with additional performance characteristics for further study.

The term telecom time synchronous clock (T-TSC) refers to a device consisting of a timeReceiver-only ordinary clock (OC) as defined in [IEEE 1588] and this Recommendation, with additional performance characteristics defined in the Annex C of [ITU-T G.8273.2].

The term primary reference time clock (PRTC) refers to the clock defined in [ITU-T G.8272]. The term enhanced primary reference time clock (ePRTC) refers to an enhanced version of the PRTC, which is being studied.

Within this Recommendation, some requirements are stated as requiring compliance with [IEEE 1588]. For implementations based on [IEEE 1588-2008], the reference to [IEEE 1588] means compliance with [IEEE 1588-2008]. For implementations based on [IEEE 1588-2019], the reference to [IEEE 1588] means compliance with [IEEE 1588-2019]. Some of these references to [IEEE 1588] include a specific clause number. In these cases, the clause number is the same in both [IEEE 1588-2008] and [IEEE 1588-2019]. If the requirements are in different clauses in the two versions of [IEEE 1588], then the text of this Recommendation shall include the specific clause for [IEEE 1588-2008] and the specific clause for [IEEE 1588-2019].

In this Recommendation the abbreviation TR stands for timeReceiver, TimeReceiver, or TIME_RECEIVER, and the abbreviation TT stands for timeTransmitter, TimeTransmitter, or TIME_TRANSMITTER.

6 Use of PTP for phase/time distribution

The [IEEE 1588] standard defines the precision time protocol designed to enable accurate time transfer. It introduces the concept of "profile", whereby aspects of the protocol may be selected and specified for a particular application.

A PTP profile has been defined in [ITU-T G.8265.1] to address applications requiring frequency synchronization only. This Recommendation defines another PTP profile, for telecom applications requiring accurate phase and time synchronization. It supports the specific architecture described in [ITU-T G.8274] in order to allow the distribution of phase/time with full timing support from the network and is based on [IEEE 1588].

In order to claim compliance with the telecom profile, the requirements of this Recommendation and the relevant requirements of [IEEE 1588], as referenced in Annex A of this Recommendation, must be met.

The detailed aspects related to the telecom profile are described in the following clauses, while the profile itself is contained in Annex A of this Recommendation. It follows the general rules for profile specification developed in [IEEE 1588].

This PTP telecom profile defines the parameters from [IEEE 1588] to be used in order to guarantee protocol interoperability between implementations and specifies the optional features, default values of configurable attributes and mechanisms that must be supported. However, it does not guarantee that the performance requirements of a given application will be met. These performance aspects are defined in other ITU-T Recommendations and imply additional elements beyond the content of the PTP profile itself.

6.1 High-level design requirements

The requirements for PTP profiles are specified in [IEEE 1588].

For operation in a telecom network, some additional criteria are also required to be consistent with standard telecom synchronization practices. Therefore, the PTP profile for time and phase distribution must meet the following high-level requirements:

1) Mechanisms must be specified to allow interoperability between the various phase/time clocks belonging to the architecture defined in [ITU-T G.8274] and described in [ITU-T G.8273].
2) Mechanisms must permit consistent operation over managed wide area telecom networks.
3) Packet-based mechanisms must allow the synchronization network to be designed and configured in a fixed arrangement.
4) Protection schemes used by packet-based systems must be based on standard telecom operational practice and allow telecom time synchronous clocks the ability to take phase and time from multiple geographically separate telecom grandmaster clocks.
5) Phase/time reference source selection based on received phase/time traceability and local priority, as well as automatic establishment of the phase/time synchronization network topology, should be permitted.

6.2 PTP modes and options

6.2.1 PTP domains

A domain consists of a logical grouping of clocks communicating with each other using the PTP protocol.

PTP domains are used to partition a network within an administrative entity. The PTP messages and datasets are associated with a domain and therefore, the PTP protocol is independent for different domains.

In this PTP telecom profile, the default PTP domain number is 24, and the range of applicable PTP domain numbers is {24 – 43}.

NOTE – This range has been selected from the user-defined PTP domain number range defined in [IEEE 1588]. Although non-overlapping ranges have been considered for the different PTP telecom profiles

so that interactions between the profiles are prevented, nothing precludes another industry from using the same user-defined PTP domain number range when defining a non-telecom PTP profile. It is the responsibility of the network operator to identify if the risk of unintentional interactions between PTP profiles exists, and to take the necessary actions to prevent such behaviour.

6.2.2 PTP messages used in the profile

This PTP profile uses the messages: *Sync*, *Follow_Up*, *Announce*, *Delay_Req*, and *Delay_Resp*.

A *Signalling* message used for reporting the time/phase offset between the local time reference and PTP is specified in Annex L of this Recommendation. The use of *Management* messages is for further study.

Signalling messages are handled according to clause 13.12 of [IEEE 1588].

Pdelay_Req, *Pdelay_Resp*, and *Pdelay_Resp_Follow_Up* messages are not used.

6.2.3 Types of PTP clocks supported in the profile

The OC, boundary clock (BC) and TC according to [IEEE 1588] are used in this profile.

There are two types of OCs:

1) An ordinary clock (OC) that can only be a grandmaster (T-GM according to the architecture defined in [ITU-T G.8274], and as included in [ITU-T G.8272]).
2) An OC that can only be a timeReceiver, i.e., a timeReceiver-only OC (T-TSC according to the architecture defined in [ITU-T G.8274], and compliant with [ITU-T G.8273.2]).

There are two types of BCs:

1) A BC that can only be a grandmaster (T-GM according to the architecture defined in [ITU-T G.8274], and as included in [ITU-T G.8272]).
2) A BC that can become a grandmaster and can also be synchronized from another PTP clock (T-BC according to the architecture defined in [ITU-T G.8274] and compliant with [ITU-T G.8273.2]).

NOTE − T-GM and GM are different concepts; GM is a status defined in [IEEE 1588] that a PTP clock may obtain if it wins the best timeTransmitter clock algorithm (BTCA), while T-GM is a type of clock defined in the architecture in [ITU-T G.8274].

The transparent clock used in this profile (T-TC according to the architecture defined in [ITU-T G.8274] and compliant with [ITU-T G.8273.3]) is the end-to-end transparent clock defined in [IEEE 1588]. It is not permitted to use peer-to-peer transparent clocks in this profile.

The support of optical transport network (OTN) ports on a T-TC is for further study.

The support of metro transport network (MTN) ports as defined in [ITU-T G.8312] on a T-TC is for further study.

The mapping between these PTP clockTypes and the phase/time clocks defined in the architecture of [ITU-T G.8274] is described in Table 1.

Table 1 – Mapping between [ITU-T G.8275.1] deviceTypes and PTP clockTypes

deviceType from [ITU-T G.8275.1]	Description	clockType from [IEEE 1588]
T-GM	TimeTransmitter-only ordinary clock (timeTransmitter with a single PTP port, always a GM, cannot be synchronized from another PTP clock)	OC
	TimeTransmitter-only boundary clock (timeTransmitter with multiple PTP ports, always a GM, cannot be synchronized from another PTP clock)	BC
T-BC	Boundary clock (may become a GM, or may be synchronized from another PTP clock)	BC
T-TSC	TimeReceiver-only ordinary clock (always a timeReceiver, cannot become a GM)	OC
T-TC	Transparent clock	End-to-end TC

6.2.3.1 Dataset member for PTP clocks supported in the profile

A management system can configure and/or read the deviceType of a PTP clock; this allows the management node to configure other properties that are specific to that type of clock implemented in the equipment that is being deployed in the network. Equipment may support different types of clocks, and therefore deviceType is a configurable attribute.

The data type for deviceType is Enumeration8, using the enumerated value per Table 2.

Table 2 – deviceType enumeration value

deviceType from [ITU-T G.8275.1]	Enumeration value (hex)
T-GM	0x10
T-BC	0x11
T-TSC	0x12
T-TC	0x13

6.2.4 One-way versus two-way operation

PTP operation must be two-way in this profile in order to transport phase/time synchronization, because propagation delay must be measured. Therefore, only two-way mode is allowed in this profile.

6.2.5 One-step versus two-step clock mode

Both one-step and two-step clocks are supported in the profile. A clock compliant with the profile may use either a one-step clock or a two-step clock.

To be compliant with [IEEE 1588], a timeReceiver port must be capable of receiving and processing messages from both one-step clocks and two-step clocks, without any particular configuration.

For IEEE 1588-2008-based implementations, the status of one-step or two-step behaviour is reported per PTP instance via defaultDS.twoStepFlag. For IEEE 1588-2019-based implementations, it is reported per PTP port via the portDS.twoStepFlag.

6.2.6 Ethernet multicast addressing for PTP messages

For the PTP profile specified in this Recommendation, when using the PTP mapping defined in Annex F of [IEEE 1588-2008] or Annex E of [IEEE 1588-2019], Ethernet multicast addressing is used for the Ethernet frames that carry the PTP messages. Both the non-forwardable multicast address 01-80-C2-00-00-0E and the forwardable multicast address 01-1B-19-00-00-00 are supported.

A T-GM, T-BC, T-TSC or T-TC clock compliant with this profile must be capable of handling both the non-forwardable multicast address 01-80-C2-00-00-0E and the forwardable multicast address 01-1B-19-00-00-00 on all its PTP-capable Ethernet ports.

For a T-GM, T-BC and T-TSC clock, the choice of the multicast address is made by configuration on a per-port basis; all the PTP messages of an Ethernet port must use the configured address for transmitting PTP messages to the remote PTP port. In the case where the remote PTP port is configured with the other address, the local PTP port must accept and process the received messages.

The default address depends on the operator policy. See the information in Appendix III of this Recommendation.

On transmission, the default mode for a T-TC that supports only Ethernet transport does not require any configuration: the PTP messages retransmitted by the T-TC must use the same multicast destination address as the received PTP message. It is mandatory to support this default mode in transmission. The default mode of transmission at an Ethernet port for a T-TC that supports both Ethernet and non-Ethernet ports is for further study.

NOTE – This profile uses Annex F of [IEEE 1588-2008], or Annex E of [IEEE 1588-2019] for the transport layer when the transport layer is Ethernet. In particular, according to the relevant Ethernet bridge models, the appropriate media access control (MAC) address of the transmit Ethernet port is placed into the source address field of the header of the Ethernet frame encapsulating the PTP packets sent by any of the PTP clocks defined in this profile (i.e., T-GM, T-BC, T-TC or T-TSC). Refer to Appendix XVII of this Recommendation for additional recommendation when the forwardable MAC address is used.

6.2.7 PTP mapping

This PTP telecom profile is based on the PTP mapping defined in

a) Annex F of [IEEE 1588-2008], or Annex E of [IEEE 1588-2019],

b) Annex H of this Recommendation, *Transport of PTP over OTN,*

c) Annex I of this Recommendation, *Transport of PTP over MTN*, and

d) Annex J of this Recommendation, *Transport of PTP over FlexE links*.

Therefore, a PTP clock compliant with the profile described in this Recommendation must be compliant with [IEEE 1588] for ports that use Ethernet transport, with Annex H of this Recommendation for ports that use OTN transport, with Annex I of this Recommendation for ports that use MTN, and with Annex J of this Recommendation for ports that use Flex Ethernet (FlexE). This profile processes PTP messages with:

a) Ethernet multicast encapsulation when the transport layer is Ethernet;

b) Either:

   i) Generic framing procedure-framed (GFP-F) encapsulation and insertion into the optical transport unit, level k (OTUk) OTN synchronization messaging channel (OSMC) or Flex OTN (FlexO) OTN synchronization messaging channel (OSMC), or

   ii) Vendor-specific encapsulation into the optical supervisory channel (OSC) OSMC,

   when the transport layer is OTN;

   NOTE − OSC OSMC does not allow interoperability between different vendors. The test of this function is for further study.

c) Ethernet multicast encapsulation, and insertion into the MTN overhead per clause 9.2.5 of [ITU-T G.8312], when the transport layer is MTN;

d) Ethernet multicast encapsulation, and insertion into the Synchronization Messaging Channel per clause 7.3.5 of [OIF FlexE IA], when the transport layer is FlexE.

PTP messages with other encapsulations should be forwarded based on the forwarding rules of the respective transport.

The transportSpecific field is used in this profile and must be set to "0".

In the scenarios currently considered, e.g., based on full timing support with T-BC and T-TC, the insertion of a IEEE 802.1Q virtual local area network (VLAN) tag or multiple VLAN tags in the frames carrying PTP messages over Ethernet is not allowed for T-GMs, T-BCs and T-TSCs (in this context, "VLAN tag" means a tag whose format is specified in clause 9 of [IEEE 802.1Q] as the sequence of octets formed by the following triplet, irrespective of the values within these octets: tag protocol identifier, tag control information, and any attendant information.). In this case, when a T-GM, T-BC or T-TSC receives a PTP message within a frame containing a VLAN tag or multiple VLAN tags, this PTP message must be discarded by the T-GM, T-BC, or T-TSC.

NOTE – For some implementations of previous editions of this Recommendation, T-GMs, T-BCs and T-TSCs may handle or discard priority-tagged frames (as defined in clause 3.190 of [IEEE 802.1Q]) carrying PTP messages over Ethernet.

Some specific T-TC based configurations are discussed in Appendix I of this Recommendation.

Using IEEE 802.1Q VLAN tags in other scenarios is for further study.

6.2.8 Message rates

Within the scope of the profile, the following messages can be used, and the corresponding indicated nominal rates shall be respected:

– *Sync* messages (if used, *Follow_up* messages will have the same rate) – nominal rate: 16 packets-per-second.

– *Delay_Req*/*Delay_Resp* messages – nominal rate: 16 packets per second.

– *Announce* messages – nominal rate: 8 packets per second.

Inter-message interval requirements shall also be respected for the transmission of *Sync* (clause 9.5.9 of [IEEE 1588]) and *Announce* (clause 9.5.8 of [IEEE 1588]) messages. In addition, the time between successive *Sync* messages shall not exceed twice the mean *Sync* interval specified above, and the time between successive *Announce* messages shall not exceed twice the mean *Announce* interval specified above.

The transmission of *Delay_Req* messages is specified in clause 9.5.11.2 of [IEEE 1588].

In addition to clause 9.5.11.2 of [IEEE 1588-2008] (first and second dashed items) or of [IEEE 1588-2019] (items (a) and (b)), a clock compliant with this profile shall follow one of the following options:

– Transmission time requirements according to clause 9.5.11.2 of [IEEE 1588-2008] (third dashed item) or of [IEEE 1588-2019] (item (c), subitem (1)), using an implementation-specific distribution. In this case, the PTP node shall, with 90% confidence, issue *Delay_Req* messages with inter-message intervals within ±30% of $2^{\text{logMinDelayReqInterval}}$ seconds.

– Transmission time requirements specified in clause 9.5.11.2 of [IEEE 1588-2008] (fourth dashed item) or of [IEEE 1588-2019] (item (c), subitem (2)).

In addition, the time between successive *Delay_Req* messages shall not exceed $2^{\text{logMinDelayReqInterval+1}}$ seconds.

As per clause 9.5.12 of [IEEE 1588], the TimeTransmitter sets the logMessageInterval in the header of Delay_Resp messages to a value accepted by the TimeTransmitter. In this profile that value is −4 (16 messages per second).

Additional background information concerning the *Delay_Req* message transmission specified in clause 9.5.11.2 of [IEEE 1588] is included in Appendix II of this Recommendation.

The rate for the *Signalling* message used for the reporting time/phase offset between the local time reference and PTP is specified in Annex L of this Recommendation.

The use of Management messages is for further study.

6.2.9 PTP clockMode (optional)

The dataset member defaultDS.clockMode represents the synchronization mode of the clock. This attribute is dynamic. Appendix VIII of [ITU-T G.8275] describes this attribute and its usage in detail.

Table 3 provides the enumeration values for clockMode.

Table 3 – clockMode enumeration values

clockMode	Enumeration value
Free-Run	0x00
Acquiring	0x01
Locked	0x02
Holdover-within-specification	0x03
Holdover-out-of-specification	0x04

NOTE 1 – The expected combinations for clock mode, PortState and clockClass are described in Table 6, as well as in Appendix VIII of [ITU-T G.8275].

The dataset member defaultDS.timeSinceCurrentClockMode indicates the time elapsed, in seconds, in the current clock mode.

The dataset member is dynamic.

The data type is UInteger32.

NOTE 2 – There is a potential loss of information due to indeterminate latency in collecting defaultDS.timeSinceCurrentClockMode.

6.2.10 PTP nodeClass (optional)

The dataset member defaultDS.nodeClass represents the clock node's time error performance compliance whether it is a PRTC (as per [ITU-T G.8272]), ePRTC (as per [ITU-T G.8272.1]), coherent network primary reference time clock (cnPRTC) (as per [ITU-T G.8272.2]), T-BC or T-TSC (as per [ITU-T G.8273.2]) or T-TC (as per [ITU-T G.8273.3]). The clock node's time error performance is determined by the worst accuracy of the ports in use (e.g., if via some ports the overall accuracy can meet class C, but over some ports it is only possible to meet class B, the overall clock node's time error performance is class B). This attribute can be used to determine the clock's expected performance, such as the maximum time error (maxTE), dynamic time error (dTE) and constant time error (cTE) metrics, as defined in the corresponding clock specifications. Where the same enumeration value is used, the deviceType value can be used to distinguish between the different clocks.

It is possible to deduce the cTE, dTE and maxTE of the clock using both the deviceType and nodeClass dataset members.

For example, if deviceType is 0x11 (T-BC) and nodeClass is 0x41, then it is a telecom boundary clock with class A compliance with associated cTE, dTE and max|TE|.

Table 4 provides the enumeration values for nodeClass.

Table 4 – nodeClass enumeration values

nodeClass	deviceType	Enumeration value	Description
This range is reserved for future clock types defined by ITU-T		0x00-0x10	Reserved for ITU-T use
PRTC-A	As per Table 2 (PRTC+T-GM)	0x11	Clock performance compliance with [ITU-T G.8272]
PRTC-B	As per Table 2 (PRTC+T-GM)	0x12	Clock performance compliance with [ITU-T G.8272]
This range is reserved for future clock types defined by ITU-T		0x13-0x20	Reserved for ITU-T use
ePRTC-A	As per Table 2 (ePRTC+T-GM)	0x21	Clock performance compliance with [ITU-T G.8272.1]
This range is reserved for future clock types defined by ITU-T		0x22-0x30	Reserved for ITU-T use
cnPRTC	As per Table 2 (cnPRTC+T-GM)	0x31	Clock performance compliance with [ITU-T G.8272.2]
This range is reserved for future clock types defined by ITU-T		0x32-0x40	Reserved for ITU-T use
Class-A	As per Table 2 (T-BC, T-TSC, T-TC)	0x41	Clock performance compliance with [ITU-T G.8273.2]/[ITU-T G.8273.3]
Class-B	As per Table 2 (T-BC, T-TSC, T-TC)	0x42	Clock performance compliance with [ITU-T G.8273.2]/[ITU-T G.8273.3]
Class-C	As per Table 2 (T-BC, T-TSC, T-TC)	0x43	Clock performance compliance with [ITU-T G.8273.2]/[ITU-T G.8273.3]
Class-D	As per Table 2 (T-BC, T-TSC)	0x44	Clock performance compliance with [ITU-T G.8273.2]
This range is reserved for future clock types defined by ITU-T		0x45-0x60	Reserved for ITU-T use
This range can be assigned by other standards development organizations (SDOs)		0x61-0xFF	Reserved for other SDOs' use

6.3 Protection aspects and Alternate BTCA

6.3.1 Alternate BTCA

The PTP profile specified in this Recommendation uses an Alternate BTCA, as described in clause 9.3.1 of [IEEE 1588]. This Alternate BTCA differs from the default BTCA of [IEEE 1588] as follows:

a) This profile uses the per-port Boolean attribute timeTransmitterOnly as specified in Annex E of [ITU-T G.8275].

b) The profile uses the per-port Boolean attribute notTimeTransmitter as specified in Annex E of [ITU-T G.8275].

c) The profile allows for multiple clocks to be active GMs simultaneously (clocks with clockClass less than 128 cannot be a timeReceiver). If there are multiple active GMs, every clock that is not a GM is synchronized by a single GM in the PTP domain.

d) The per-port attribute localPriority is assigned to each port r of a clock and is used in the determination of E_{rbest} and E_{best}. Each parent clock or foreign timeTransmitter clock dataset, whose *Announce* information was received on the port r, is appended with the localPriority attribute of the local port r before the dataset comparison defined in Figure 2 and Figure 3 below is invoked. The localPriority attribute is not transmitted in *Announce* messages. This attribute is used as a tie-breaker in the dataset comparison algorithm, in the event that all other previous attributes of the datasets being compared are equal. The localPriority attribute is set via the configurable, unsigned integer, port dataset member portDS.localPriority. The data type for this attribute is UInteger8. The range of values for this attribute is {1-255}. The default value for this attribute is 128. A clock compliant with this PTP profile is allowed to support a subset of the values defined in the range.

e) The attribute localPriority is assigned to the local clock, to be used if needed when the data associated with the local clock, D_0, is compared with data on another potential GM received via an *Announce* message. The local clock localPriority attribute is set via the configurable, unsigned integer, default dataset member defaultDS.localPriority. The data type for this attribute is UInteger8. The range of values for this attribute is {1-255}. The default value for this attribute is 128. A clock compliant with this PTP profile is allowed to support a subset of the values defined in the range.

f) The dataset comparison algorithm is modified according to Figures 2 and 3 in clause 6.3.7.

NOTE 1 − For a T-BC, the ports for which the timeTransmitterOnly attribute is FALSE should be selected according to the network synchronization plan. One typical use case where this parameter should remain TRUE is to prevent timing from propagating from the access portion of the network to the core portion of the network.

NOTE 2 − The use of timeTransmitterOnly is intended primarily to be used in two scenarios:

1) A PTP port of a T-GM;
2) A PTP port of a T-BC that is facing the 'downstream' direction towards the access portion of a tree topology.

The use of the timeTransmitterOnly parameter in other scenarios, such as on PTP ports participating in a ring architecture, may result in unintended operation, especially during reconfiguration or topology changes.

6.3.2 Considerations on the use of the localPriority attributes

The localPriority attributes provide a powerful tool in defining the synchronization network architecture.

The use of the default values for these attributes as defined by the Alternate BTCA results in a timing-loop free synchronization network.

Proper planning will be mandatory to avoid timing-loops when configuring values different from the default ones.

6.3.3 Static clock attribute priority1

In this PTP profile, the clock attribute priority1 is static. It is initialized to a default value equal to the midpoint value, 128, of its range, and this value must not be changed.

The priority1 parameter is not used in this version of the PTP telecom profile. Future versions may consider using this attribute, this is for further study.

6.3.4 Clock attribute priority2

In this PTP profile, the clock attribute priority2 is configurable.

It is initialized to a default value, equal for T-GM and T-BC clocks to the midpoint value, 128, of its range {0-255}. The default value for T-TSC clocks is 255, and the range is {255}.

A T-GM or T-BC compliant with this PTP profile must support all the values of priority2 defined in the range. A T-TSC compliant with this profile must support, on reception, all the values of priority2 defined in the full [IEEE 1588] range (i.e., {0-255}).

Appendix IV of this Recommendation describes possible use cases for the priority2 attribute; Appendix X of this Recommendation describes possible use cases for the priority2 attribute under Note 3 of Table 6; other cases are for further study.

6.3.5 Other clock attributes

A PTP clock compliant with this PTP profile must support all the values of clockClass, clockAccuracy, and offsetScaledLogVariance upon reception as [shall not discard] defined in the full [IEEE 1588] range.

The applicable values of the clock attribute clockClass are specified in clause 6.4.

NOTE 1 – It is not expected that equipment compliant with this profile and deployed in an ITU-T G.8275.1 network will receive clockClass values not specified in Table 6 or Table 8. If a clockClass value not specified in Table 6 or Table 8 is received, then the equipment may raise an implementation specific alarm. Future revisions of this profile may include clockClass values not defined in Table 6 or Table 8.

The following values of the clock attribute clockAccuracy (defaultDS.clockQuality.clockAccuracy) apply for the following situations:

– 0x20 for a T-GM connected to an ePRTC in locked-mode (i.e., ePRTC traceable to global navigation satellite system (GNSS));
– 0x21 for a T-GM connected to a PRTC-A or PRTC-B in locked-mode (i.e., PRTC traceable to GNSS) or a T-GM connected to an ePRTC where the ePRTC is in phase/time holdover within [ITU-T G.8272.1] ePRTC-A specification as specified in Table 3 of [ITU-T G.8272.1];
– 0xFE for a T-GM not connected to an ePRTC nor a PRTC in locked-mode, or a T-GM connected to an ePRTC in phase/time holdover but outside the holdover specification in Table 3 of [ITU-T G.8272.1];
– 0xFE for a T-BC, all the time.

The following values of the clock attribute offsetScaledLogVariance (defaultDS.clockQuality.offsetScaledLogVariance) apply for the following situations (see computation details in Appendix XIV of [ITU-T G.8275]):

– 0x448D for a T-GM connected to an ePRTC in locked-mode (i.e., ePRTC traceable to GNSS) or in holdover-within-specification mode. This corresponds to a time deviation (TDEV) of 1 ns, at observation interval of 10 000 seconds;

– 0x4932 for a T-GM connected to a PRTC-B in locked-mode (i.e., PRTC-B traceable to GNSS). This corresponds to a TDEV of 5 ns, at observation interval of 10 000 seconds.
– 0x4E5D for a T-GM connected to a PRTC-A in locked-mode (i.e., PRTC-A traceable to GNSS). This corresponds to a TDEV of 30 ns, at observation interval of 10 000 seconds;
– 0xFFFF for a T-GM not connected to an ePRTC or a PRTC in locked-mode, or a T-GM connected to an ePRTC in phase/time holdover but outside the holdover specification in Table 3 of [ITU-T G.8272.1];
– 0xFFFF for a T-BC, all the time.

NOTE 2 – An implementation may use other values of clockAccuracy and offsetScaledLogVariance as defined in [IEEE 1588].

The optional configuration parameters defaultDS.forcedClockQuality.clockAccuracy and defaultDS.forcedClockQuality.offsetScaledLogVariance, are used to force the values in Locked mode. The default values for PRTC-A/PRTC-B/ePRTC are implementation specific, but the recommended values are 0x21/0x21/0x20 respectively for clockAccuracy and 0x4E5D/0x4932/0x448D respectively for offsetScaledLogVariance.

Configuring the clockAccuracy and offsetScaledLogVariance on PTP clocks of a live network may change the current PTP network topology, careful design should be considered. For further details, refer to Appendix XV of [ITU-T G.8275].

6.3.6 State decision algorithm

The state decision algorithm applicable to the Alternate BTCA of the PTP profile specified in this Recommendation is given in Figure 1. After a decision is reached by the use of this algorithm, the datasets of the local clock are updated as specified in clause 9.3.5 of [IEEE 1588]. Details on the use of the algorithm are given in clause 9.3.3 of [IEEE 1588].

6.3.7 Data set comparison algorithm

The dataset comparison algorithm for the Alternate BTCA of the PTP profile specified in this Recommendation is given in Figures 2 and 3 below. With this algorithm, one clock is compared with another using the datasets representing those clocks, appended with the localPriority attribute. Details on the use of the algorithm are given in clause 9.3.4 of [IEEE 1588].

If either of the datasets, A or B, in Figures 2 and 3 contain the data of the parent clock or a foreign timeTransmitter clock, the corresponding localPriority for its dataset is the localPriority of the local port r on which the information from that parent clock or foreign timeTransmitter clock has been received (see item (d) of clause 6.3.1).

If either of the datasets, A or B, in Figures 2 and 3 contain the data of the local clock, D_0, the corresponding localPriority for that dataset is the localPriority of the local clock (see item (e) of clause 6.3.1).

NOTE 1 – It is recommended that the entire dataset comparison algorithm described in Figures 2 and 3 be implemented even if some parameters are currently static, because they may be used in future versions of this Recommendation.

NOTE 2 – The block in Figure 2 "GM clockClass of A is 127 or less" enables different T-BCs in the network to be synchronized by different T-GMs when multiple T-GMs are deployed.

NOTE 3 – If portDS.SF is TRUE on port r, then the PTP port should set the respective E_{rbest} to the empty set. As a result, the computation of E_{best} will not use the information contained in any Announce messages received on the port r. Signal fail (SF) is described in clause 6.3.9.

NOTE 4 – Procedures related to defaultDS.timeReceiverOnly are defined in clauses 9.2.2 of [IEEE 1588-2008] or 9.2.2.1 of [IEEE 1588-2019], and 9.3 of [IEEE 1588].

NOTE 5 – Procedures related to portDS.timeTransmitterOnly are defined in clause 9.2.2.2 of [IEEE 1588-2019] and Annex E of [ITU-T G.8275].

NOTE 6 – Procedures related to portDS.notTimeTransmitter are defined in Annex E of [ITU-T G.8275].

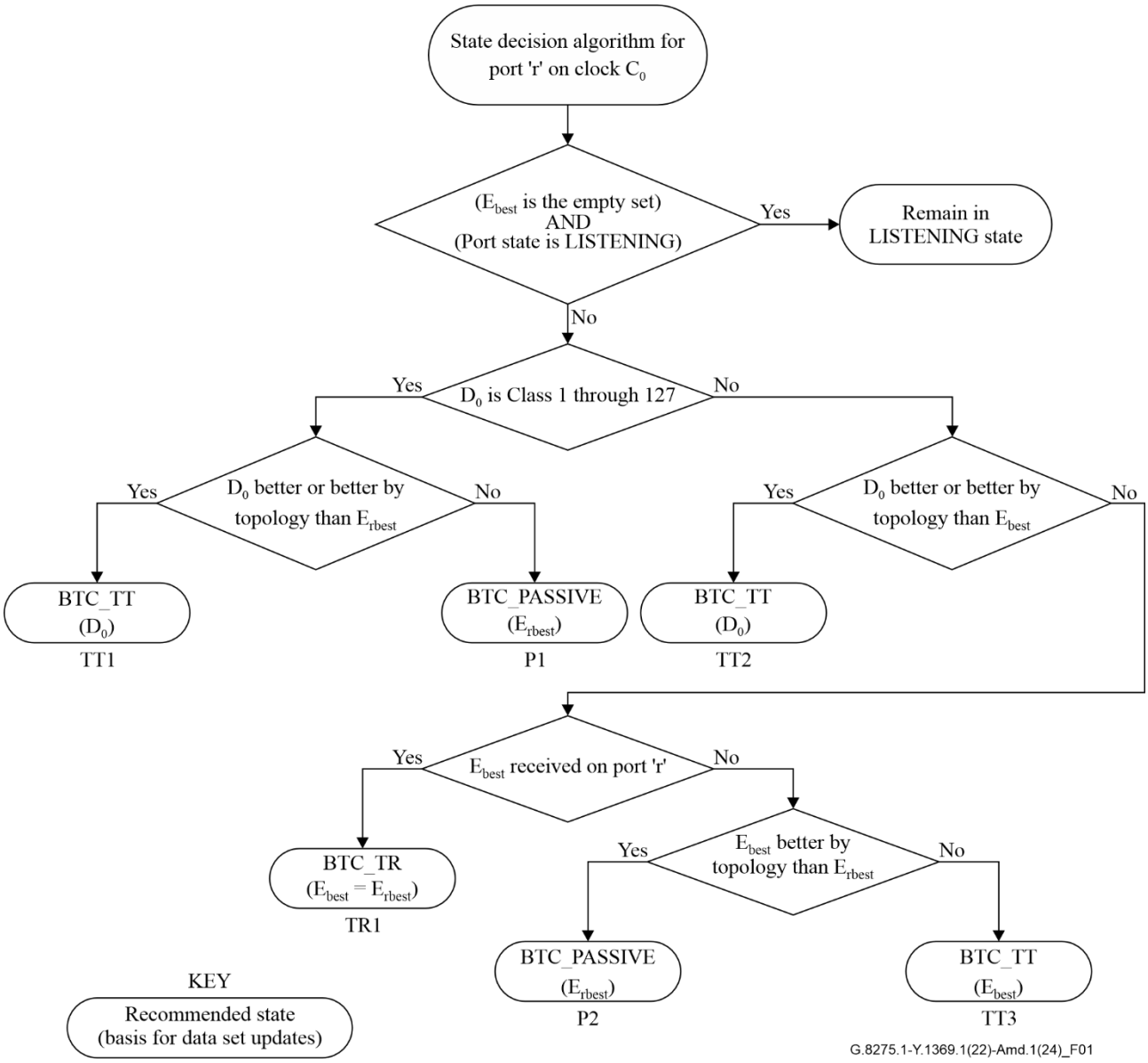


Figure 1 – State decision algorithm for Alternate BTCA

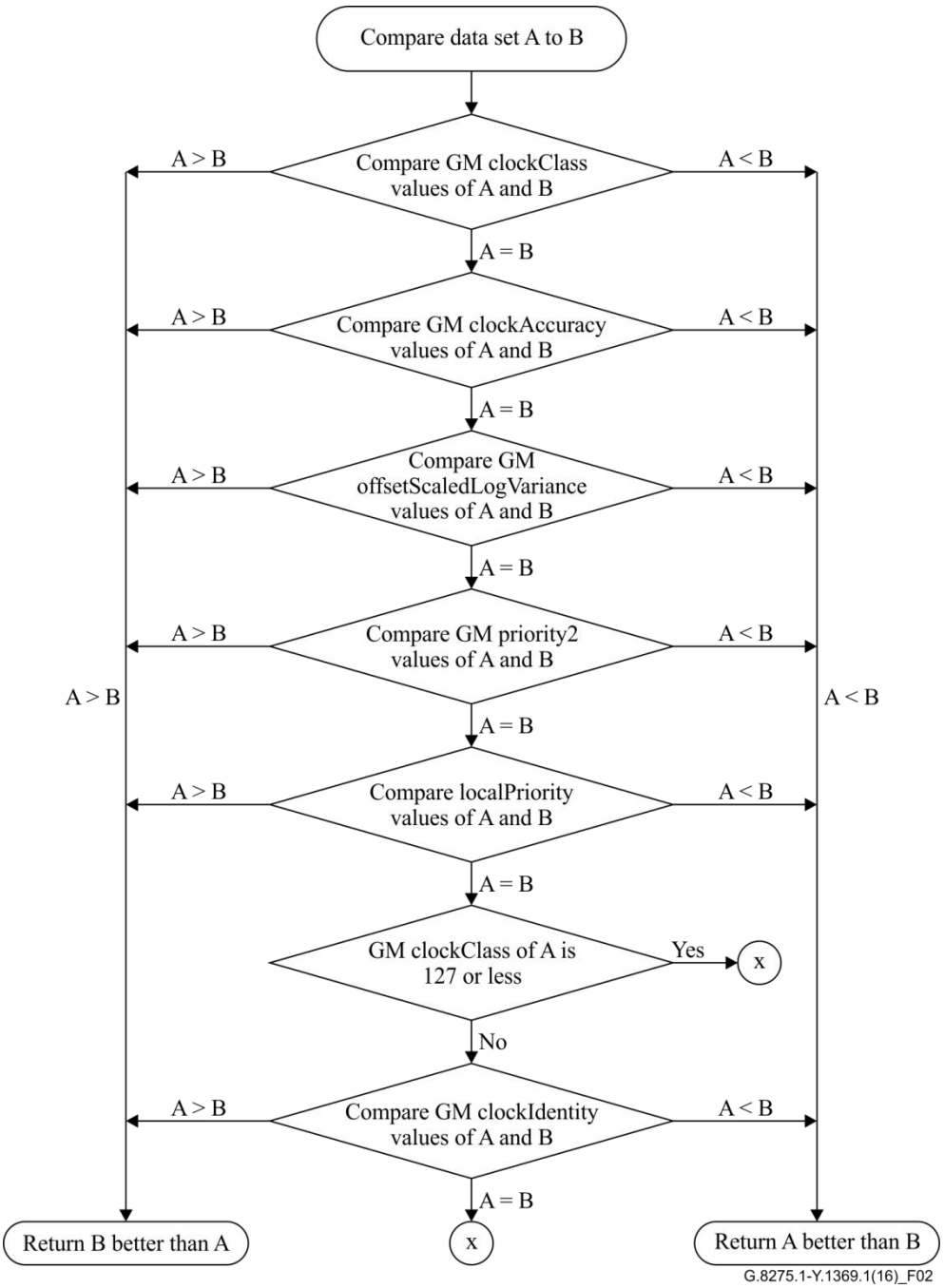


Figure 2 – Data set comparison algorithm, part 1, for Alternate BTCA

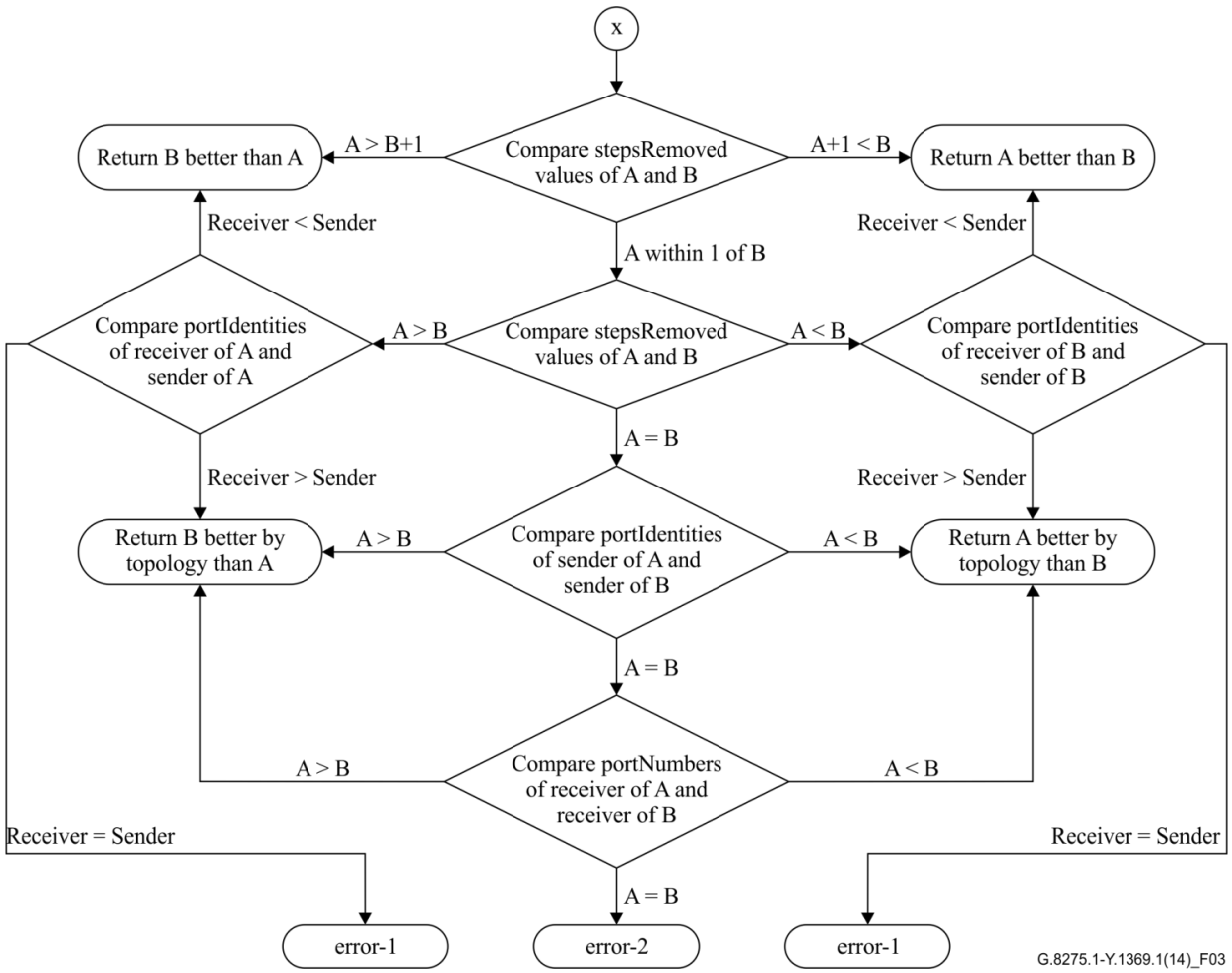


Figure 3 – Data set comparison algorithm, part 2, for Alternate BTCA

6.3.8 Unused PTP fields

Some PTP fields are not used in this PTP profile. This clause defines the actions applicable to these unused PTP fields.

Table A.8 in clause A.10 of this Recommendation defines the PTP common header flag values, and whether or not each flag is used in this profile.

In addition, the following fields are not used in this profile:

– The "controlField" in the common header of PTP messages is not used in this profile. This field must be ignored by the receiver for all types of PTP messages;

– The "priority1" field in the *Announce* message is not used and must be set to a fixed value specified in clause 6.3.3.

When a PTP clock receives a PTP message with a field, whose use is not specified in this PTP profile, containing a value outside the allowed range, then this field of the PTP message must be ignored, without discarding the PTP message.

As an example, a PTP clock compliant with this PTP profile must ignore on reception the field value for the following fields. A clock compliant with this PTP profile must not update its local datasets with the ingress value for these fields:

– flagField – unicastFlag;

– flagField – PTP profile Specific 1;

– flagField – PTP profile Specific 2.

When a PTP clock receives a PTP message with a field, whose use is specified in this PTP profile, containing a value outside the allowed range for reception, then this entire PTP message must be discarded. The allowed range for a parameter on reception is the same as the range for the corresponding default dataset parameter, except for the attributes clockClass, clockAccuracy, offsetScaledLogVariance, and priority2 (see clauses 6.3.4 and 6.3.5).

As an example, a compliant clock must discard on reception the ingress packet (General and Event messages) when any of the following fields are outside the allowed range for the profile:

– domainNumber;

– versionPTP.

The clock's local dataset must not be updated with the ingress value.

NOTE 1 – If a clock receives an *Announce* message with the "priority1" field set to a value other than 128, and if the clock advertising this value is selected as the GM, then 128 is re-advertised by the receiving clock. The unused attribute priority1 is ignored by the receiving clock for the purpose of the Alternate BTCA.

NOTE 2 – The allowed ranges for reception for the clock attributes priority2, clockClass, clockAccuracy, and offsetScaledLogVariance are the respective full [IEEE 1588] ranges, see clauses 6.3.4 and 6.3.5.

6.3.9 Packet timing signal fail

This clause is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein. An implementation may support all, none, or a subset of these types of packet timing signal fail (PTSF).

This clause defines the notion of PTSF, which corresponds to a signal indicating a failure of the PTP packet timing signal received by a port.

The format used for PTSF notification is implementation-specific (e.g., event-based, state variable, etc.).

Three types of PTSF may be raised (and cleared) in a PTP clock:

1) packet timing signal fail - loss of timing messages (PTSF-LOTM), lack of reception of PTP timing messages from a peer port (loss of the packet timing signal): as long as a port of a PTP clock no longer receives the expected ingress timing messages sent by a peer PTP port (i.e., Sync and subsequently Follow_Up and Delay_Resp messages when Delay_Req messages are sent), then a PTSF-LOTM associated with this port must occur. A timeout period for reception of Sync messages or Delay_Resp messages (i.e., syncReceiptTimeout and delayRespReceiptTimeout) for these timing messages must be implemented in the PTP port before triggering the PTSF-LOTM (the range and default value of these timeout parameters are defined in Table A.5).

   The value of syncReceiptTimeout shall specify the number of Sync message intervals that have to pass without receipt of a Sync and, if the twoStep flag of the Sync message is TRUE, a Follow_Up message before the triggering of the PTSF-LOTM.

   The value of delayRespReceiptTimeout shall specify the number of Delay_Req message intervals that have to pass without receipt of a Delay_Resp message before the triggering of the PTSF-LOTM, if Delay_Req messages are sent on the same port.

   When the PTP port enters the UNCALIBRATED or PASSIVE state, or when port monitoring is enabled on this port as per Annex G or K of this Recommendation, a timer-based mechanism should be activated to detect the potential loss and recovery of the expected ingress messages. PTSF-LOTM is therefore set when one or more type of expected messages are determined to be lost (as described above) and cleared if none of the expected messages are determined to be lost.

2) PTSF-unusable, unusable PTP packet timing signal received by the timeReceiver or passive port of a PTP clock, exceeding the input tolerance of the PTP clock (noisy packet timing signal): as long as the PTP packet timing signal is not usable for the PTP clock to achieve the performance target (e.g., excessive noise or abnormal PTP timestamps), then a PTSF-unusable associated with this timeTransmitter must occur. The criteria used to determine that the packet timing signal is not suitable for use or not are for further study.

3) Packet timing signal fail – synchronization uncertain (short name PTSF-SU): uncertain timing signal received by the timeReceiver or passive port of a PTP clock, as long as the synchronizationUncertain flag of an Announce message received from an upstream clock is TRUE, a PTSF-SU associated with this timeTransmitter must occur.

When the beginning or the end of a PTSF condition occurs, the clock may update the PTP portDS.SF value and generate a state decision event, which triggers the running of the Alternate BTCA. As described in clause 6.3.7 a value of portDS.SF of TRUE can be used to exclude PTP ports from the Alternate BTCA selection process. An implementation may set the PTP portDS.SF to TRUE for only a subset of the PTSF listed above; that is, the implementation may set portDS.SF to TRUE for some PTSF types and leave portDS.SF as FALSE for others. For example, when PTSF-SU is raised, if portDS.SF is not set to TRUE, then this would allow the PTP topology to be built as described in Annex D of [ITU-T G.8275].

When the beginning or the end of a PTSF condition occurs, the clock may update the values of the additional dataset members related to portDS.SF (i.e., portDS.SF-LOTM, portDS.SF-unusable and portDS.SF-SU, for PTSF-LOTM, PTSF unusable and PTSF-SU respectively).

NOTE 1 – See Appendix XIV of this Recommendation for more discussion of deployments with transparent clocks.

NOTE 2 – Annex K of this Recommendation includes an option for dynamic monitoring by a PTP port when its portDS.SF has been set to TRUE due to PTSF.

NOTE 3 – If a port reports the loss of Announce messages using the ANNOUNCE_RECEIPT_TIMEOUT event, then it may force portDS.SF to FALSE to avoid redundant alarms.

PTSF for a virtual PTP port is implementation specific. However, the behaviour of PTSF for a virtual PTP port needs to be consistent with the behaviour of PTSF for a PTP port.

6.3.10 Wait-to-restore time

This clause is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

The wait-to-restore (WTR) time ensures that a previously degraded PTP port or a previously degraded input virtual PTP port is not considered as a candidate reference (with normal handling of PTP messages) until portDS.SF has been cleared for a certain time. Afterwards, the Alternate BTCA can use Announce messages received on that port.

The wait-to-restore time is configurable in the range of 0 to 12 minutes, in steps of 1 minute, for all PTP ports. The default value is 5 minutes. Configuring an immediate expiration of a running WTR timer is possible using a zero value for defaultDS.wtrTime. For more details on reconfiguring a timer, refer to Appendix XVI of [ITU-T G.8275].

If the WTR timer is active (meaning there is a WTR event on a port, and the remaining WTR time is greater than zero), then the port is in the WTR state. A port dataset member portDS.wtrState indicates whether a PTP port or an input virtual PTP port is in a WTR state. It is defined as follows:

– FALSE: the port is not in a WTR state;

– TRUE: the port is in a WTR state.

A port dataset member portDs.wtrRemainingTime indicates the number of seconds remaining before the WTR timer expires. When portDS.wtrState changes from TRUE to FALSE, portDS.wtrRemainingTime must be 0.

Separate wait-to-restore timers are used for each PTP port and each input virtual PTP port. Each wait-to-restore timer can be cleared individually.

NOTE – Using wait-to-restore time related to the ANNOUNCE_RECEIPT_TIMEOUT_EXPIRES event is for further study.

6.4 Phase/time and frequency traceability information

To deliver phase/time and frequency traceability information to the downstream nodes (e.g., using PTP or 1 PPS+ToD interface), the clockClass values and traceability flags described in Table 6 must be used in this PTP telecom profile.

When a PTP clock is locked to an upstream PTP parent clock, the traceability information of this parent clock is used to update the timePropertiesDS that will be passed to the downstream nodes, as per PTP. This means that the attributes and flags in the PTP header or 1 PPS+ToD Time event message will always reflect the phase/time and frequency traceability information from the current parent clock, regardless of the frequency traceability of the T-BC's physical layer. Otherwise, when the PTP clock is not locked to an upstream PTP clock, so that the PTP clock timePropertiesDS is populated based on the status of the local clock, the frequencyTraceable flag present in the header of the Announce PTP messages on the TimeTransmitter ports is defined in this profile as follows: if the PTP clock

- is a T-GM that is traceable to a PRTC in locked mode or
- is a T-GM or a T-BC in holdover or free-run mode that has a timescale traceable to a primary reference clock (PRC),

then this parameter must be set to TRUE, otherwise it must be FALSE. This flag is not used in the Alternate BTCA defined in clause 6.3; the values provided for this flag in Table 6 can be used by a network operator for monitoring purposes or by end applications to take definitive action as described in Appendix VIII of this Recommendation.

When a T-GM first enters holdover, it may enter holdover-within-specification mode and downgrade the clockClass value that it uses to 7. It then calculates if the time error at its output is still within the holdover specification. When the T-GM determines that the time error at its output has exceeded the holdover specification, it enters holdover-out-of-specification mode and downgrades the clockClass value that it uses to 140, 150 or 160 depending on the quality of its frequency reference (internal oscillator or received physical layer frequency signal on an external interface).

When a T-BC first enters holdover, it may enter holdover-within-specification mode and downgrade the clockClass value that it uses to 135. It then calculates if the time error at its output is still within the holdover specification. When the T-BC determines that the time error at its output has exceeded the holdover specification, it enters holdover-out-of-specification mode and downgrades the clockClass value that it uses to 165 (internal oscillator or received physical layer frequency signal on an external interface).

The criteria that a T-GM or T-BC uses to determine the limits for the holdover-within-specification mode can be either a configurable timer (refer to Table A.1 for defaultDS.holdoverWithinSpecificationDuration) or a configurable time error threshold (see defaultDS.holdoverWithinSpecificationTimeErrorThreshold in Table A.1).

defaultDS.holdoverWithinSpecificationDuration is an optional parameter used to configure the duration of the holdover-within-specification mode in minutes. The default value is zero, and its range is {0–65535}; the implementation may support a smaller maximum value.

defaultDS.holdoverWithinSpecificationTimeErrorThreshold is an optional parameter used to configure the time error (TE) threshold of the holdover-within-specification mode in nanoseconds. The default value is zero, and its range is {0–65535}; the implementation may support a smaller maximum value.

Configuring an immediate downgrade to the holdover-out-of-specification mode is possible using a zero value for defaultDS.holdoverWithinSpecificationDuration or defaultDS.holdoverWithinSpecificationTimeErrorThreshold, either before or during holdover. For more details on reconfiguring a timer in the context of the use of defaultDS.holdoverWithinSpecificationDuration, refer to Appendix XVI of [ITU-T G.8275].

A static parameter (see defaultDS.holdoverWithinSpecificationCapability in Table A.1) indicates the supported mode. defaultDS.holdoverWithinSpecificationCapability is an optional parameter to indicate the mode supported by the equipment per Table 5.

Table 5 – defaultDS.holdoverWithinSpecificationCapability enumeration value

defaultDS.holdoverWithinSpecificationCapability	**Enumeration value (hex)**
Does not support any of the modes	0x00
Supports only holdoverWithinSpecificationDuration mode	0x01
Supports only holdoverWithinSpecificationTimeErrorThreshold mode	0x02
Supports both modes	0x03

If both modes are supported, the configuration for both defaultDS.holdoverWithinSpecificationDuration and defaultDS.holdoverWithinSpecificationTimeErrorThreshold is mutually exclusive; they cannot be configured simultaneously.

The optional estimation of the remaining duration for the holdover-within-specification mode, in minutes, (refer to Table A.1 for the dynamic defaultDS.holdoverWithinSpecificationRemainingDuration parameter) is available during locked mode and holdover-within-specification mode.

NOTE 1 – The applicable holdover specification depends on the design and budgeting of the synchronization network. See Appendix V of [ITU-T G.8271.1] for examples of network budgeting. A typical value for the holdover budget, described in the failure scenario (b) depicted in [ITU-T G.8271.1] Table V.1, when using the T-GM or T-BC for holdover while still meeting a total time error of 1.5 µs, is 400 ns applicable to T-BC or T-TSC class A and B deployments. Other values detailed in failure scenario (d) depicted in [ITU-T G.8271.1] Table V.2, are applicable to T-BC or T-TSC class C deployments.

NOTE 2 – When the term clockClass is used with respect to the property of the individual PTP clock (T-GM, T-BC, T-TSC) it is referring to dataset member defaultDS.clockQuality.clockClass.

NOTE 3 – Holdover-within-specification and holdover-out-of-specification modes are defined in [ITU-T G.8275] clause VIII.2.

Table 6 – Applicable clockClass values

Phase/time traceability description	defaultDS. clockQuality. clockClass	frequencyTraceable flag	timeTraceable flag
T-GM connected to a PRTC in locked mode (e.g., PRTC traceable to GNSS)	6	TRUE	TRUE
T-GM in holdover-within-specification mode, traceable to Category 1 frequency source (Note 1)	7	TRUE	TRUE
T-GM in holdover-within-specification mode, non-traceable to Category 1 frequency source (Note 1)	7	FALSE	TRUE
T-BC in holdover-within-specification mode, traceable to Category 1 frequency source (Note 1)	135	TRUE	TRUE
T-BC in holdover-within-specification mode, non-traceable to Category 1 frequency source (Note 1, Note 3)	135	FALSE	TRUE
T-GM in holdover-out-of-specification mode, traceable to Category 1 frequency source (Note 1, Note 3)	140	TRUE	FALSE
T-GM in holdover-out-of-specification mode, traceable to Category 2 frequency source (Note 1, Note 3)	150	FALSE	FALSE
T-GM in holdover-out-of-specification mode, traceable to Category 3 frequency source (Note 1, Note 3)	160	FALSE	FALSE
T-BC in holdover-out-of-specification mode (Note 1, Note 3)	165	(Note 2)	FALSE
T-GM or T-BC in free-run mode	248	(Note 2)	FALSE
T-TSC	255	(Note 2)	As per PTP

NOTE 1 – Either the configurable timer or the configurable time error threshold could be set to zero so that the T-GM or T-BC would enter holdover out of holdover specification mode, and advertise a degraded clockClass value directly after losing traceability to a PRTC, regardless of the time error on the output ports of the clock.

NOTE 2 – The frequencyTraceable flag may be TRUE or FALSE, depending on the availability of a PRC-traceable physical layer frequency input signal.

NOTE 3 – As an option, the clockClass range of a T-BC can be extended from (135, 165, 248) to (135, 140, 150, 160, 165, 248) for some cases, where (a) 140, 150, 160, and 165 are related to the quality of the frequency reference, (b) the applicable circumstances of 140, 150, and 160 are the same as for the T-GM, and (c) 165 corresponds to Synchronous Equipment clock (SEC). Details are in Appendix X of this Recommendation. If this option is used, then in a single PTP domain, all PTP clocks should implement this option (and should not be intermixed with clocks that do not implement this option). Details are for further study.

NOTE 4 – The term "holdover" in this table refers to "time holdover".

NOTE 5 – Refer to Appendix VII of this Recommendation for more information about the behaviour of a T-BC that was previously synchronizing to a T-GM or a T-BC that is advertising clock class values corresponding to 'holdover-within-holdover specification', 'holdover-out-of-holdover specification', or 'free-run mode'.

NOTE 6 – A T-BC that is locked to a PRTC through virtual PTP port (refer to Figure II.1 in [ITU-T G.8274], Appendix II) and meets the holdover performance requirements of a T-GM, may follow the clockClass assignments of a T-GM if the T-BC enters holdover (a) due to the loss of the connectivity to that PRTC, and (b) no other PTP source is available.

NOTE 7 – Refer to Appendix XVI of this Recommendation for more information on use-cases and the use of clockClass and priority2.

Table 7 describes how the clock quality levels (QLs) defined in [ITU-T G.781] are mapped to Category 1, 2 and 3 frequency sources used in Table 6.

Table 7 – Mapping of [ITU-T G.781] clock QLs to Category 1, 2, 3 frequency sources

Category (in Table 6 and Table 8)	ITU-T G.781 option I QLs	ITU-T G.781 option II QLs
Category 1 frequency source	Quality level (QL)-ePRTC, QL-PRTC, QL-ePRC, QL-PRC	QL-ePRTC, QL-PRTC, QL-ePRC, QL-primary reference source (PRS)
Category 2 frequency source	QL- primary level synchronization supply unit (SSU-A)	QL-stratum 2 (ST2)
Category 3 frequency source	QL-secondary level SSU (SSU-B)	QL-stratum 3 enhanced (ST3E)
NOTE – Other frequency source categories, while not used in Table 6, are possible. An example is a category containing QL-synchronous equipment clock (SEC) and QL- stratum 3 (ST3).		

NOTE 4 – The case of a T-BC acting as a GM, with an external phase/time input coming from a PRTC, is handled by means of a virtual PTP port with associated E_{rbest} attributes as described in Annex C of this Recommendation. The general case of a T-BC with a phase/time external synchronization input different from a PRTC is for further study.

Table 8 presents a subset of the clockClass values of Table 6 based on the quality of the frequency reference, and the mapping of the corresponding values used by some equipment deployed prior to this Recommendation.

NOTE 5 – When interoperability with equipment deployed prior to this Recommendation is needed, both sets of clockClass values would need to be supported. Other aspects may be required for full interoperability.

Table 8 – clockClass values for equipment deployed prior to this Recommendation

Phase/time traceability description	Values defined in Table 6	Values prior to this Rec.
T-GM connected to a PRTC in locked mode (e.g., PRTC traceable to GNSS)	6	6
T-GM in holdover-out-of-specification mode, traceable to Category 1 frequency source (Note 1)	140	7
T-GM in holdover-out-of-specification mode, traceable to Category 2 frequency source (Note 1)	150	(Note 2)
T-GM in holdover-out-of-specification mode, traceable to Category 3 frequency source (Note 1)	160	52
T-BC in holdover-out-of-specification mode, using unspecified frequency source (Note 1)	165	187
T-TSC	255	255
NOTE 1 – Initially after advertising clockClass values greater than 6, a clock may still be within the holdover specification. NOTE 2 – Refer to the applicable value specified for the equipment. NOTE 3 – The term "holdover" in this table refers to "time holdover".		

7 ITU-T PTP profile for phase/time distribution with full timing support from the network

The PTP profile for phase/time distribution with full timing support from the network is contained in Annex A of this Recommendation.

8 Security aspects

Appendix XIII of [ITU-T G.8275] lists techniques that can be used to enhance the security of PTP and that are compatible with this profile.

Annex A

ITU-T PTP profile for phase/time distribution with full timing support from the network

(This annex forms an integral part of this Recommendation.)

This annex contains the PTP telecom profile for phase/time distribution with full timing support from the network, as required by [IEEE 1588]. In order to claim compliance with this PTP telecom profile, the requirements in this annex and in the body of this Recommendation must both be met.

A.1 Profile identification

profileName: ITU-T PTP profile for phase/time distribution with full timing support from the network

profileVersion: 2.10 (for an implementation based on [IEEE 1588-2008])

profileVersion: 2.11 (for an implementation based on [IEEE 1588-2019])

profileIdentifier: 00-19-A7-01-02-0A (for an implementation based on [IEEE 1588-2008])

profileIdentifier: 00-19-A7-01-02-0B (for an implementation based on [IEEE 1588-2019])

See clause A.11 for details of compatibility between profile versions.

This profile is specified by ITU-T.

A copy may be obtained from www.itu.int.

A.2 PTP attribute values

The default values and ranges of the PTP attributes for use in this profile are contained in Tables A.1, A.2, A.3, A.4, A.5, A.6 and A.7. For the attributes clockClass, clockAccuracy, offsetScaledLogVariance, and priority2, the ranges shown are those for the defaultDS.

NOTE 1 – A boundary clock follows the rules of [IEEE 1588] for the selection of the parent clock, the updating of parentDS, and the transmission of Announce messages, so it may transmit values different from the defaultDS values.

These tables provide a default initialization value and range for each dataset member for:

– Telecom grandmaster: ordinary clock or boundary clock that can only act as a GM (T-GM according to [ITU-T G.8274] – first PTP clock of the chain);

– Telecom time synchronous clock: ordinary clock with a clockClass = 255 (T-TSC according to [ITU-T G.8274] – last PTP clock of the chain);

– Telecom boundary clock: boundary clock that may or may not be a GM; such a clock will be a GM if it is the best clock in the network (T-BC according to [ITU-T G.8274] – intermediate PTP clocks of the chain);

– Telecom transparent clock: end-to-end transparent clock (T-TC according to [ITU-T G.8274] – intermediate PTP clock of the chain).

The mapping between these PTP clockTypes and the phase/time clocks defined in the architecture of [ITU-T G.8274] is described in Table 1, clause 6.2.3.

The attributes defined in Tables A.6 and A.7 are only required for implementations based on [IEEE 1588-2008]; they have been deprecated in [IEEE 1588-2019]. For implementations based on [IEEE 1588-2008], these transparent clock datasets are not used for telecom grandmasters, telecom time synchronous clocks and telecom boundary clocks. For implementations based on

[IEEE 1588-2008], they are mandatory for telecom transparent clocks. For implementations based on [IEEE 1588-2019], these transparent clock datasets are optional.

NOTE 2 – A feature that is deprecated by [IEEE 1588], is no longer recommended or required.

Some attributes in these tables are associated with optional features of this Recommendation. Therefore, these attributes are also optional, unless the associated feature is implemented, in which case these attributes must be supported.

Unless explicitly documented within these tables, the data type of a dataset member is as per PTP.

[IEEE 1588] defines other attributes that are not specified in Tables A.1, A.2, A.3, A.4, A.5, A.6 and A.7. Attributes not specified by this profile can use the IEEE 1588 default initialization values and ranges.

Table A.1 – defaultDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
Clause	**Data type**	**Clause**	**Data type**		**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**
8.2.1.2.1	As per PTP	(Note 3)	(Note 3)	defaultDS. twoStepFlag (static)	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}
8.2.1.2.2	As per PTP	8.2.1.2.2	As per PTP	defaultDS. clockIdentity (static) (Note 2)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.1.2.3	As per PTP	8.2.1.2.3	As per PTP	defaultDS. numberPorts (dynamic) (Note 2)	1 for OC As per PTP for BC	{1} for OC As per PTP for BC	1	{1}	As per PTP	As per PTP
8.2.1.3.1.1	As per PTP	8.2.1.3.1.2	As per PTP	defaultDS.clock Quality. clockClass (dynamic)	248	{6, 7, 140, 150, 160, 248}	255	{255}	248	{135, 165, 248}
8.2.1.3.1.2	As per PTP	8.2.1.3.1.3	As per PTP	defaultDS.clock Quality.clock Accuracy (dynamic)	0xFE	As per PTP (Note 1)	0xFE	{0xFE}	0xFE	{0xFE}
8.2.1.3.1.3	As per PTP	8.2.1.3.1.4	As per PTP	defaultDS. clockQuality. offsetScaledLog Variance (dynamic)	0xFFFF	As per PTP (Note 1)	0xFFFF	{0xFFFF}	0xFFFF	{0xFFFF}
8.2.1.4.1	As per PTP	8.2.1.4.1	As per PTP	defaultDS. priority1 (configurable)	128	{128}	128	{128}	128	{128}
8.2.1.4.2	As per PTP	8.2.1.4.2	As per PTP	defaultDS. priority2 (configurable)	128	{0-255}	255	{255}	128	{0-255}

Table A.1 – defaultDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
Clause	Data type	Clause	Data type		Default initialization value (Note 5)	Range	Default initialization value (Note 5)	Range	Default initialization value (Note 5)	Range
8.2.1.4.3	As per PTP	8.2.1.4.3	As per PTP	defaultDS.domainNumber (configurable) (Note 2)	24	{24-43}	24	{24-43}	24	{24-43}
8.2.1.4.4	As per PTP	8.2.1.4.4	As per PTP	defaultDS.timeReceiverOnly (configurable)	FALSE	{FALSE}	TRUE	{TRUE}	FALSE	{FALSE}
(Note 4)	As per PTP	8.2.1.4.5	As per PTP	defaultDS.sdoId (configurable) (Note 2)	0x000	0x000	0x000	0x000	0x000	0x000
New member	UInteger8	New member	UInteger8	defaultDS.localPriority (configurable)	128	{1-255}	128	{1-255}	128	{1-255}
New member	UInteger8	8.2.1.5.4	As per PTP	defaultDS.maxStepsRemoved (configurable)	255	{2-255}	255	{2-255}	255	{2-255}
New member	Enume-ration8	New member	Enume-ration8	defaultDS. deviceType (configurable)	(Note 6) (Note 8)	(Note 6) (Note 8)	(Note 6) (Note 8)	(Note 6) (Note 8)	(Note 6) (Note 8)	(Note 6) (Note 8)
New member	Octet[6]	New member	Octet[6]	defaultDS.profileIdentifier (configurable)	(Note 7)	(Note 7)	(Note 7)	(Note 7)	(Note 7)	(Note 7)
New member	Boolean	8.2.1.5.2	As per PTP	defaultDS. instanceEnable (configurable)	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}
New member	UInteger16	New member	UInteger16	defaultDS.holdoverWithinSpecificationDuration (configurable) (optional)	0 minutes	{0-65535 minutes} (Note 9)	Not used	Not used	0 minutes	{0-65535 minutes} (Note 9)
New member	UInteger16	New member	UInteger16	defaultDS.holdoverWithinSpecificationTimeErrorThreshold (configurable) (optional)	0 ns	{0-65535 ns} (Note 9)	Not used	Not used	0 ns	{0-65535 ns} (Note 9)
New member	Enumeration8	New member	Enumeration8	defaultDS.holdoverWithinSpecificationCapability (static) (optional)	0	See clause 6.4	Not used	Not used	0	See clause 6.4
New member	Uinteger16	New member	Uinteger16	defaultDS.holdoverWithinSpecificationRemainingDuration (dynamic) (optional)	0 minutes	{0-65535 minutes}	Not used	Not used	0 minutes	{0-65535 minutes}
New member	UInteger8	New member	UInteger8	defaultDS. wtrTime (configurable) (optional)	Not used	Not used	5 minutes	{0-12 minutes, in step of 1 minute}	5 minutes	{0-12 minutes, in step of 1 minute}

Table A.1 – defaultDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
Clause	**Data type**	**Clause**	**Data type**		**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**
New member	Enumeration8	New member	Enumeration8	defaultDS.clockMode (dynamic) (optional)	Free-Run mode	Refer Table 3 in clause 6.2.9	Free-Run mode	Refer Table 3 in clause 6.2.9	Free-Run mode	Refer Table 3 in clause 6.2.9
New member	Enumeration8	New member	Enumeration8	defaultDS.nodeClass (dynamic) (Note 2) (optional)	as per compliance	0x11, 0x12, 0x21, 0x31	as per compliance	{0x41-0x44}	as per compliance	{0x41-0x44}
New member	UInteger32	New member	UInteger32	defaultDS.timeSinceCurrentClockMode (dynamic) (optional)	0 second	{0 to 4294967295 seconds}	0 second	{0 to 4294967295 seconds}	0 second	{0 to 4294967295 seconds}
New member	Boolean	New member	Boolean	defaultDS.withLTR (dynamic) (optional)	Not used	Not used	Implementation specific	{FALSE, TRUE}	Not used	Not used
New member	Boolean	New member	Boolean	defaultDS.LTRValid (dynamic) (optional)	Not used	Not used	Implementation specific	{FALSE, TRUE}	Not used	Not used
New member	Boolean	New member	Boolean	defaultDS.timeStatus (dynamic) (optional)	Not used	Not used	Implementation specific	{FALSE, TRUE}	Not used	Not used
New member	Boolean	New member	Boolean	defaultDS.ptpReference (dynamic) (optional)	Not used	Not used	Implementation specific	{FALSE, TRUE}	Not used	Not used
New member	UInteger16	New member	UInteger16	defaultDS.messageInterval (configurable) (optional)	Not used	Not used	60	Implementation specific	Not used	Not used
New member	(Note 10)	New member	(Note 10)	defaultDS.forcedClockQuality.clockAccuracy (configurable) (optional)	(Note 1)	(Note 10)	0xFE	{0xFE}	0xFE	{0xFE}
New member	(Note 11)	New member	(Note 11)	defaultDS.forcedClockQuality.offsetScaledLogVariance (configurable) (optional)	(Note 1)	(Note 11)	0xFFFF	{0xFFFF}	0xFFFF	{0xFFFF}

Table A.1 – defaultDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
Clause	**Data type**	**Clause**	**Data type**		**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**
New member	UInteger8	New member	UInteger8	defaultDS.clockClassAvailabilityThreshold (configurable) (optional)	6	{0-255}	6	{0-255}	6	{0-255}

NOTE 1 – Examples of applicable values are shown in clause 6.3.5. Additional values, not specified in clause 6.3.5, may be applicable to future use cases. These additional values are for further study.

NOTE 2 – Applicable also to T-TC in case of an IEEE 1588-2019-based implementation.

NOTE 3 – In case of IEEE 1588-2019-based implementations this dataset member is deprecated.

NOTE 4 – Applicable to IEEE 1588-2019-based implementations; this dataset member does not exist in [IEEE 1588-2008].

NOTE 5 – If a default value is not provided by this Recommendation, then it is implementation-specific.

NOTE 6 – This attribute can be used to set the deviceType in use in the PTP clock instance. See clause 6.2.3.1 for more information and applicable values.

NOTE 7 – This dataset member is set according to the profileIdentifier specified in clause A.1.

NOTE 8 – [IEEE 1588-2019] has defined a dataset member with a similar name (i.e., defaultDS.instanceType) that is used for a different purpose.

NOTE 9 – The full range is not expected to be supported by an implementation, and the implementation may support a smaller maximum value. An implementation should be verified only over the expected operating conditions.

NOTE 10 – defaultDS.forcedClockQuality.clockAccuracy uses the same DataType and range as defaultDS.clockQuality.clockAccuracy.

NOTE 11 – defaultDS.forcedClockQuality.offsetScaledLogVariance uses the same DataType and range as defaultDS.clockQuality.offsetScaledLogVariance.

Table A.2 – currentDS dataset member specifications

Clause from [IEEE 1588-2008]	Clause from [IEEE 1588-2019]	Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
			Default initialization value (Note 3)	**Range**	**Default initialization value (Note 3)**	**Range**	**Default initialization value (Note 3)**	**Range**
8.2.2.2	8.2.2.2	currentDS.stepsRemoved (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.2.3	8.2.2.3	currentDS.offsetFromTimeTransmitter (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.2.4	(Note 1)	currentDS.meanPathDelay (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
(Note 2)	8.2.2.4	currentDS.meanDelay (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
New member	New member	currentDS.offsetFromPtpToLTR (dynamic) (optional)	Not used	Not used	Implementation specific	Implementation specific	Not used	Not used

NOTE 1 – In case of IEEE 1588-2019-based implementations, this dataset member is deprecated.

NOTE 2 – Applicable to IEEE 1588-2019-based implementations; this dataset member does not exist in [IEEE 1588-2008].

NOTE 3 – If a default value is not provided by this Recommendation, then it is implementation-specific.

Table A.3 – parentDS dataset member specifications

Clause from [IEEE 1588-2008]	Clause from [IEEE 1588-2019]	Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
			Default initialization value (Note 1)	Range	Default initialization value (Note 1)	Range	Default initialization value (Note 1)	Range
8.2.3.2	8.2.3.2	parentDS.parentPortIdentity (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.3.6	8.2.3.6	parentDS.grandmasterIdentity (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.3.7	8.2.3.7	parentDS.grandmasterClockQuality (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.3.8	8.2.3.8	parentDS.grandmasterPriority1 (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.3.9	8.2.3.9	parentDS.grandmasterPriority2 (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
NOTE – If a default value is not provided by this Recommendation, then it is implementation specific.								

Table A.4 – timePropertiesDS dataset member specifications

Clause from [IEEE 1588-2008]	Clause from [IEEE 1588-2019]	Members of the dataset	Telecom grandmaster requirements		Telecom time synchronous clock requirements		Telecom boundary clock requirements	
			Default initialization value (Note 2)	Range	Default initialization value (Note 2)	Range	Default initialization value (Note 2)	Range
8.2.4.2	8.2.4.2	timePropertiesDS.currentUtcOffset (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.4.3	8.2.4.3	timePropertiesDS.currentUtcOffsetValid (dynamic)	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}
8.2.4.4	8.2.4.4	timePropertiesDS.leap59 (dynamic)	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}
8.2.4.5	8.2.4.5	timePropertiesDS.leap61 (dynamic)	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}
8.2.4.6	8.2.4.6	timePropertiesDS.timeTraceable (dynamic)	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}	FALSE	{FALSE, TRUE}
8.2.4.7	8.2.4.7	timePropertiesDS.frequencyTraceable (dynamic)	FALSE	{FALSE, TRUE} (Note 1)	FALSE	{FALSE, TRUE} (Note 1)	FALSE	{FALSE, TRUE} (Note 1)
8.2.4.8	8.2.4.8	timePropertiesDS.ptpTimescale (dynamic)	TRUE	{TRUE}	TRUE	{TRUE}	TRUE	{TRUE}
8.2.4.9	8.2.4.9	timePropertiesDS.timeSource (dynamic)	0xA0	As per PTP	0xA0	As per PTP	0xA0	As per PTP
NOTE 1 – Refer to clause 6.4 for details on when each value applies. NOTE 2 – If a default value is not provided by this Recommendation, then it is implementation specific.								

Table A.5 – portDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	TimeTransmitter port requirements of telecom grandmaster		TimeReceiver port requirements of telecom time synchronous clock		Telecom boundary clock requirements	
Clause	**Data type**	**Clause**	**Data type**		**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**
8.2.5.2.1	As per PTP	8.2.15.2.1	As per PTP	portDS.portIdentity.clockIdentity (static) (Note 2)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.5.2.1	As per PTP	8.2.15.2.1	As per PTP	portDS.portIdentity.portNumber (static) (Note 2)	1 for OC As per PTP for BC	{1} for OC As per PTP for BC	1	{1}	As per PTP	As per PTP
8.2.5.3.1	As per PTP	8.2.15.3.1	As per PTP	portDS.portState (dynamic)	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP	As per PTP
8.2.5.3.2	As per PTP	8.2.15.3.2	As per PTP	portDS.logMinDelayReqInterval (dynamic) (Note 2)	−4	{−4}	−4	{−4}	−4	{−4}
8.2.5.4.1	As per PTP	8.2.15.4.1	As per PTP	portDS.logAnnounceInterval (configurable)	−3	{−3}	−3 (Note 1)	{−3} (Note 1)	−3	{−3}
8.2.5.4.2	As per PTP	8.2.15.4.2	As per PTP	portDS.announceReceiptTimeout (configurable)	3	{3 – z} z is FFS	3	{3 – z} z is FFS	3	{3 – z} z is FFS
8.2.5.4.3	As per PTP	8.2.15.4.3	As per PTP	portDS.logSyncInterval (configurable)	−4	{−4}	−4 (Note 1)	{−4} (Note 1)	−4	{−4}
8.2.5.4.4	As per PTP	8.2.15.4.4	As per PTP	portDS.delayMechanism (configurable)	01	{01}	01	{01}	01	{01}
8.2.5.4.6	As per PTP	8.2.15.4.6	As per PTP	portDS.versionNumber (configurable)	2	{2}	2	{2}	2	{2}
(Note 3)	(Note 3)	8.2.15.4.7	As per PTP	portDS.minorVersionNumber (configurable) (Note 2)	1	{1}	1	{1}	1	{1}
(Note 3)	(Note 3)	8.2.15.4.8	As per PTP	portDS.delayAsymmetry (configurable) (Note 2)	Not used	Not used	As per PTP	As per PTP	As per PTP	As per PTP
New member	Boolean	9.2.2.2	As per PTP	portDS.timeTransmitterOnly (configurable) (Note 4)	TRUE	{TRUE, FALSE}	FALSE	{FALSE}	TRUE	{TRUE, FALSE}
New member	UInteger 8	New member	UInteger 8	portDS.local Priority (configurable)	128	{1-255}	128	{1-255}	128	{1-255}
New member	UInteger 16	New member	UInteger 16	portDS.syncReceiptTimeout (configurable)	Not used	Not used	Implementation specific	{3-65535} (Note 6)	Implementation specific	{3-65535} (Note 6)

Table A.5 – portDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	TimeTransmitter port requirements of telecom grandmaster		TimeReceiver port requirements of telecom time synchronous clock		Telecom boundary clock requirements	
Clause	Data type	Clause	Data type		Default initialization value (Note 5)	Range	Default initialization value (Note 5)	Range	Default initialization value (Note 5)	Range
New member	UInteger 16	New member	UInteger 16	portDS.delayRespReceiptTimeout (configurable)	Not used	Not used	Implementation specific	{3-65535} (Note 6)	Implementation specific	{3-65535} (Note 6)
New member	Boolean	New member	Boolean	portDS.SF (dynamic) (optional)	FALSE	{FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS.notTimeTransmitter (configurable) (Note 4)	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS.monitorSender (configurable) (optional)	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS.monitorReceiver (configurable) (optional)	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS. SF-LOTM (dynamic) (optional)	FALSE	{FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS. SF-unusable (dynamic) (optional)	FALSE	{FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	New member	Boolean	portDS.SF-SU (dynamic) (optional)	FALSE	{FALSE}	FALSE	{TRUE, FALSE}	FALSE	{TRUE, FALSE}
New member	Boolean	8.2.15.5.1	As per PTP	portDS.portEnable (configurable)	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}	As per PTP	{FALSE, TRUE}
(Note 3)	(Note 3)	New member	Boolean	portDS.twoStepFlag (static) (optional) (Note 7)	Implementation specific	{FALSE, TRUE}	Implementation specific	{FALSE, TRUE}	Implementation specific	{FALSE, TRUE}
New member	Boolean	New member	Boolean	portDS.wtrState (dynamic) (optional)	Not used	Not used	Implementation specific	{FALSE, TRUE}	Implementation specific	{FALSE, TRUE}
New member	Uinteger16	New member	Uinteger16	portDS.wtrRemainingTime (dynamic) (optional)	Not used	Not used	0 second	{0-720 seconds}	0 second	{0-720 seconds}
New member	Boolean	New member	Boolean	portDS.withLTR (dynamic) (optional)	Not used	Not used	Not used	Not used	Implementation specific	{FALSE, TRUE}
New member	Boolean	New member	Boolean	portDS.LTRValid (dynamic) (optional)	Not used	Not used	Not used	Not used	Implementation specific	{FALSE, TRUE}
New member	Boolean	New member	Boolean	portDS.timeStatus (dynamic) (optional)	Not used	Not used	Not used	Not used	Implementation specific	{FALSE, TRUE}

Table A.5 – portDS dataset member specifications

[IEEE 1588-2008]		[IEEE 1588-2019]		Members of the dataset	TimeTransmitter port requirements of telecom grandmaster		TimeReceiver port requirements of telecom time synchronous clock		Telecom boundary clock requirements	
Clause	**Data type**	**Clause**	**Data type**		**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**	**Default initialization value (Note 5)**	**Range**
New member	Boolean	New member	Boolean	portDS.ptpReference (dynamic) (optional)	Not used	Not used	Not used	Not used	Implemen-tation specific	{FALSE, TRUE}
New member	TimeInterval	New member	TimeInterval	portDS. offsetFromPtpTo LTR (dynamic) (optional)	Not used	Not used	Not used	Not used	Implemen-tation specific	Impleme n-tation specific

NOTE 1 – This type of message is not sent by a timeReceiver-only OC.

NOTE 2 – Applicable also to T-TC in case of an IEEE 1588-2019-based implementation.

NOTE 3 – Applicable to IEEE 1588-2019-based implementations; this dataset member does not exist in [IEEE 1588-2008].

NOTE 4 – Setting both portDS.timeTransmitterOnly and portDS.notTimeTransmitter with the value TRUE concurrently is not permitted.

NOTE 5 – If a default value is not provided by this Recommendation, then it is implementation specific.

NOTE 6 – The full range is not expected to be supported by an implementation. An implementation should be verified only over the expected operating conditions.

NOTE 7 – Some implementations may support a configurable twoStepFlag parameter. Some PTP applications or equipment may benefit from the use of a configurable parameter, e.g., when disabling/enabling MACsec on the interface.

Table A.6 – transparentClockDefaultDS dataset member specifications

Clause from [IEEE 1588-2008]	Members of the dataset	Telecom transparent clock requirements	
		Default initialization value (Note 1)	**Range**
8.3.2.2.1	transparentClockDefaultDS.clockIdentity (static)	As per PTP	As per PTP
8.3.2.2.2	transparentClockDefaultDS.numberPorts (dynamic)	As per PTP for TC	As per PTP for TC
8.3.2.3.1	transparentClockDefaultDS.delayMechanism (configurable)	01	{01}
8.3.2.3.2	transparentClockDefaultDS.primaryDomain (configurable)	24	{24-43}
8.3.2.2.1	transparentClockDefaultDS profileIdentifier (configurable)	(Note 2)	(Note 2)

NOTE 1 – If a default value is not provided by this Recommendation, then it is implementation-specific.

NOTE 2 – This dataset member is set according to the profileIdentifier specified in clause A.1.

NOTE – According to [IEEE 1588-2019], the transparent clock dataset members used in [IEEE 1588-2008] are deprecated. The relevant dataset members of a PTP instance should apply instead (see Tables A.1 and A.5).

Table A.7 – transparentClockPortDS dataset member specifications

Clause from [IEEE 1588-2008]	Members of the dataset	Telecom transparent clock requirements	
		Default initialization value (Note 2)	Range
8.3.3.2.1	transparentClockPortDS.portIdentity (static)	As per PTP	As per PTP
8.3.3.3.1	transparentClockPortDS.logMinPdelayReqInterval (dynamic)	(Note 1)	(Note 1)
8.3.3.3.2	transparentClockPortDS.faultyFlag (dynamic)	FALSE	{FALSE, TRUE}
8.3.3.3.3	transparentClockPortDS.peerMeanPathDelay (dynamic)	(Note 1)	(Note 1)
NOTE 1 – It is not used in this profile, and therefore equipment is not required to support it. NOTE 2 – If a default value is not provided by this Recommendation, then it is implementation specific.			

NOTE – According to [IEEE 1588-2019], the transparent clock dataset members used in [IEEE 1588-2008] are deprecated. The relevant dataset members of a PTP instance should apply instead (see Tables A.1 and A.5).

A.3 PTP options

A.3.1 Node types required, permitted, or prohibited

In this profile, the permitted node types are: ordinary clocks, boundary clocks and end-to-end transparent clocks.

In this profile, the prohibited node types are: peer-to-peer transparent clocks.

A.3.2 One-step versus two-step clock mode

Both one-step and two-step clocks are permitted. A clock must be capable of receiving and handling messages transmitted from both one-step and two-step clocks. A clock is not required to support both one-step and two-step mode for transmitting messages.

A.3.3 Transport mechanisms required, permitted, or prohibited

In this profile, the allowed transport mechanisms are:

a) Ethernet, as per Annex F of [IEEE 1588-2008] or Annex E of [IEEE 1588-2019];

b) OTN via: (i) GFP-F encapsulation as per [ITU-T G.7041], and insertion of the GFP-F frames into the OTUk OSMC as per clause 15.7.2.4 of [ITU-T G.709] or into the FlexO OSMC as per clause 9.2.10.1 of [ITU-T G.709.1], or (ii) the OSC OSMC as per clause 14.1 of [ITU-T G.709], where in this case the encapsulation is vendor-specific;

c) MTN, with PTP in Ethernet frames transported per clause 9.2.5 of [ITU-T G.8312];

d) FlexE links, with PTP in Ethernet frames transported per clause 7.3.5 of [OIF FlexE IA].

At least one of the four transport mechanisms must be supported. For transport over Ethernet, both the non-forwardable multicast address, 01-80-C2-00-00-0E, and the forwardable multicast address, 01-1B-19-00-00-00, are required to be supported for compliance with this profile.

All other transport mechanisms are for further study within the scope of this profile.

A.3.4 Unicast messages

All messages are sent using multicast, using one of the two multicast addresses in clause A.3.3. The unicast mode is not permitted in this version of the profile.

A.4 Best timeTransmitter clock algorithm options

This profile uses the Alternate BTCA described in clause 6.3.

A.5 Path delay measurement option (delay request/delay response)

The delay request/delay response mechanism is used in this profile. The peer delay mechanism must not be used in this profile.

A.6 Clock identity format

For implementations based on [IEEE 1588-2008], the procedures to use a 48-bit extended unique identifier (EUI-48) to create the EUI-64 clockIdentity as described in clause 7.5.2.2.2 of [IEEE 1588-2008] are no longer recommended. If a clockIdentity is formed by mapping an EUI-48 to an EUI-64, and if the EUI-48 was assigned from a MAC address – medium (MA-M) or MAC address – small (MA-S), it is possible that the clockIdentity will be a duplicate of a clockIdentity formed directly from a different MA-M or MA-S (i.e., by appending bits to the end of that different MA-M or MA-S). Only if the EUI-48 was formed from an organizationally unique identifier (OUI) MAC address – large (MA-L), is the uniqueness ensured. For new implementations based on [IEEE 1588-2008], the clockIdentity shall be constructed as per clause 7.5.2.2.2 of [IEEE 1588-2019]. Non-IEEE clockIdentity formats are not supported.

For implementations based on [IEEE 1588-2019], the clockIdentity shall be constructed as per clause 7.5.2.2.2 of [IEEE 1588-2019].

A.7 Configuration management options

Management aspects are for further study and will be specified in a future version of this profile.

A.8 Security aspects

For guidance on security for PTP applicable to this profile, refer to clause 8.

A.9 Other optional features of [IEEE 1588]

Other optional features of [IEEE 1588] are not used in this version of the profile.

For implementations based on [IEEE 1588-2008] these include unicast message negotiation (clause 16.1 of [IEEE 1588-2008]), alternate timescales (clause 16.3 of [IEEE 1588-2008]), grandmaster clusters (clause 17.3 of [IEEE 1588-2008]), unicast discovery (clause 17.5 of [IEEE 1588-2008]), acceptable timeTransmitter table (clause 17.6 of [IEEE 1588-2008]), and the experimental cumulative frequency scale factor offset (Annex L of [IEEE 1588-2008]).

For implementations based on [IEEE 1588-2019] these include unicast message negotiation (clause 16.1 of [IEEE 1588-2019]), alternate timescale offsets (clause 16.3 of [IEEE 1588-2019]), grandmaster clusters (clause 17.2 of [IEEE 1588-2019]), unicast discovery (clause 17.4 of [IEEE 1588-2019]), acceptable timeTransmitter table (clause 17.5 of [IEEE 1588-2019]), and the Cumulative frequency transfer method for synchronizing clocks (clause 16.10 of [IEEE 1588-2019]).

Some aspects of the optional feature of an alternate timeTransmitter are used by optional Annexes G and K of this Recommendation.

A.10 PTP common header flags

The PTP common header flag values, and whether or not each flag is used in this profile, are given in Table A.8.

NOTE – Some of these flags are used only in certain PTP messages, and not in all the PTP messages, see clause 13.3.2.6 of [IEEE 1588-2008] or clause 13.3.2.8 of [IEEE 1588-2019].

For implementations based on [IEEE 1588-2008] the following rule defined in clause 13.3.2.6 of [IEEE 1588-2008], must be respected: "For message types where the bit is not defined in Table 20, the values shall be FALSE."

For implementations based on [IEEE 1588-2019] the following rule defined in clause 13.3.2.8 of [IEEE 1588-2019], must be respected: "For message types where the bit is not defined in Table 37, the values shall be FALSE."

Table A.8 – PTP flags

Octet	Bit	Flag	Value to be sent	Behaviour for the receiving node
0	0	alternateTimeTransmitterFlag	(Note 7)	As per PTP
0	1	twoStepFlag	As per PTP	Used
0	2	unicastFlag	FALSE	Flag is ignored
0	5	PTP profile Specific 1	FALSE	Flag is ignored
0	6	PTP profile Specific 2	FALSE	Flag is ignored
0	7	Reserved	FALSE	Reserved by PTP and flag is ignored
1	0	leap61	As per PTP (Note 2)	Used
1	1	leap59	As per PTP (Note 2)	Used
1	2	currentUtcOffsetValid	As per PTP (Note 3)	Used (Notes 3, 4, 5, 6)
1	3	ptpTimescale	TRUE	Used
1	4	timeTraceable	See Table 6	Used
1	5	frequencyTraceable	See Table 6	Used
1	6	(Note 1)	(Note 1)	(Note 1)

NOTE 1 – An additional flag "synchronizationUncertain" has been defined in Annex E of this Recommendation; the use of the "synchronizationUncertain" flag is optional.

NOTE 2 – When a clock is in holdover, within holdover specification, the PTP clock may continue to advertise the last known leap second event. If there was no pending leap second event, then the PTP clock continues to advertise FALSE for the pending leap second fields (leap59 and leap61). If there was a pending leap second event, the PTP clock may choose either to advertise FALSE for the pending leap second fields immediately or to continue to advertise the leap second event. In the latter instance the PTP clock would clear the leap59 and leap61 second event field(s) and adjust the coordinated universal time (UTC) offset field at the appropriate time based on its local PTP time (i.e., the local PTP time's UTC timescale rolling over at UTC midnight) if the PTP clock is still in holdover, within holdover specification. When a clock is in holdover, out of holdover specifications, the PTP clock behaviour with respect to the leap second event is implementation-specific. It is recommended that the PTP clock continue to advertise any upcoming leap second event as appropriate.

NOTE 3 – When a clock is in holdover, within holdover specification, the PTP clock may continue to advertise the last known UTC offset with UTC offset valid TRUE. If the last known UTC offset valid was FALSE, then the PTP clock continues to advertise FALSE. If the last known UTC offset valid was TRUE then the PTP clock may choose either to advertise UTC offset valid FALSE immediately (freezing the UTC offset value) or to continue to advertise the last known UTC offset with UTC offset valid TRUE. The UTC offset field may be updated as described in Note 2 above. When a clock is in holdover, out of holdover specifications, the PTP clock behaviour with respect to the UTC offset is implementation-specific.

NOTE 4 – Usage of currentUtcOffset from an Announce message, which indicates currentUtcOffsetValid as FALSE, may lead to wrong UTC time calculation.

NOTE 5 – The Alternate BTCA in this profile does not consider or use currentUtcOffsetValid or currentUtcOffset.

NOTE 6 – The PTP clocks in this profile do not use currentUtcOffsetValid or currentUtcOffset for phase/time synchronization.

NOTE 7 – The alternateTimeTransmitterFlag is transmitted with the value of FALSE except for the cases specified in Annex G of this Recommendation.

A.11 Profile version compatibility

A specific profile version may have backward compatibility with earlier versions of this profile. Table A.9 lists which earlier versions are compatible with each version. Compatibility means that the nodes can operate in the same network provided optional features in a specific profile that are not present in the earlier version are not enabled.

Table A.9 – profileVersion compatibility

profileVersion	Base IEEE version	Created in ITU-T G.8275.1 Edition (and date)	Compatible with the earlier profileVersions
2.11	[IEEE 1588-2019]	v2025 (10/2025) Edition 5.0	2.10, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.0 (Note)
2.10	[IEEE 1588-2008]	v2025 (10/2025) Edition 5.0	2.8, 2.6, 2.4, 2.2, 2.1, 2.0, 1.0 (Note)
2.9	[IEEE 1588-2019]	v2022 Amd2 (08/2024) Edition 4.2	2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.0 (Note)
2.8	[IEEE 1588-2008]	v2022 Amd2 (08/2024) Edition 4.2	2.6, 2.4, 2.2, 2.1, 2.0, 1.0 (Note)
2.7	[IEEE 1588-2019]	v2022 Amd1 (01/2024) Edition 4.1	2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.0 (Note)
2.6	[IEEE 1588-2008]	v2022 Amd1 (01/2024) Edition 4.1	2.4, 2.2, 2.1, 2.0, 1.0 (Note)
2.5	[IEEE 1588-2019]	v2022 (11/2022) Edition 4.0	2.4, 2.3, 2.2, 2.1, 2.0, 1.0 (Note)
2.4	[IEEE 1588-2008]	v2022 (11/2022) Edition 4.0	2.2, 2.1, 2.0, 1.0 (Note)
2.3	[IEEE 1588-2019]	v2020 Amd2 (06/2021) Edition 3.2	2.2, 2.1, 2.0, 1.0 (Note)
2.2	[IEEE 1588-2008]	v2020 Amd1 (11/2020) Edition 3.1	2.1, 2.0, 1.0 (Note)
2.1	[IEEE 1588-2008]	v2016 Amd1 (08/2017) Edition 2.1	2.0, 1.0 (Note)
2.0	[IEEE 1588-2008]	v2016 (06/2016) Edition 2.0	1.0 (Note)
1.0	[IEEE 1588-2008]	v2014 (07/2014) Edition 1.0	
NOTE – Version 1 of this profile enforced a limited range of acceptable values for clockClass, clockAccuracy, offsetScaledLogVariance and, for the T-TSC, priority2. Reception of values outside the acceptable range caused the Announce message to be discarded. Version 2.x of the profile supports the full range of these attributes as defined by PTP. In networks deploying ePRTCs, which use new values of clockAccuracy and offsetScaledLogVariance that are outside the version 1 range, all clocks need to use version 2.x of the profile. If no ePRTCs are to be deployed in the network, then the network can operate with a mixture of version 1 and version 2.x clocks.			

A.12 Other optional features used in this profile

Annex F of [ITU-T G.8275] provides performance monitoring functionality that can be used in this profile.

Annex B

Options to establish the PTP topology with the Alternate BTCA

(This annex forms an integral part of this Recommendation.)

Options to establish the PTP topology with the Alternate BTCA are described in Annex C of [ITU-T G.8275].

Annex C

Inclusion of an external phase/time input interface in a T-BC

(This annex forms an integral part of this Recommendation.)

The model for inclusion of a unidirectional, external phase/time interface on a PTP clock is provided in Annex B of [ITU-T G.8275].

Annex D

Path trace (optional)

(This annex forms an integral part of this Recommendation.)

In order to track the actual path of the PTP synchronization reference in the network, the Path trace option as per clause 16.2 in [IEEE 1588] can be optionally supported by the PTP clock compliant with this profile.

This feature is optional; however, if it is supported it is necessary that it should be supported as described below.

As an example, this option can be used to support troubleshooting analysis in case of failures in the network.

It should be possible to configure a PTP clock so that the Path trace type length value (TLV) is not further forwarded by that clock (e.g., this could be required at the network administration interfaces).

NOTE 1 – There might be cases where not all PTP clocks in the network support the Path trace TLV. The expected behaviour is that in the case of an incoming Announce message carrying the Path trace TLV, this TLV is dropped by these nodes.

NOTE 2 – T-TCs in the path could add their own clockIdentity to the Path trace TLV.

Annex E

Synchronization uncertain indication (optional)

(This annex forms an integral part of this Recommendation.)

The synchronization uncertain indication is described in Annex D of [ITU-T G.8275].

Annex F

Use of stepsRemoved to limit reference chain (optional)

(This annex forms an integral part of this Recommendation.)

The parameter stepsRemoved defined in [IEEE 1588] is used to indicate the distance between a clock and the grandmaster clock in the network. It was originally intended to be used as a helpful tool to detect cyclic paths that may occur in PTP systems. By default, it is required that a clock does not qualify any Announce messages where the stepsRemoved field is equal to, or greater than, 255 in order to ensure that rogue frames are extinguished.

This profile includes a default dataset member, maxStepsRemoved, that allows the operator to configure the value to less than 255. The value configured is typically the same in all clocks in the PTP domain. When configured, a clock would not qualify any Announce messages received where the stepsRemoved field in the ingress header matches or exceedes the maxStepsRemoved field. There would be two primary use cases for this configurability.

First, referring to [ITU-T G.8271.1] and [ITU-T G.8274] analysis has been performed to show that up to 20 clocks may be traversed between the grandmaster and a timeReceiver while maintaining the proper network performance. If the operator wanted to ensure the network performance limit was not exceeded, or that the chain length was not exceeded, the operator may configure the value to be smaller (such as 20 or 21).

Second, if the operator does deploy PTP in a ring topology, the operator may configure the maxStepsRemoved parameter to a smaller value so that the clocks may more quickly identify rogue frames and take corrective action to extinguish them and update the topology.

Annex G

Monitoring alternate timeTransmitter time information provided by a peer PTP port (optional)

(This annex forms an integral part of this Recommendation.)

This annex is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

A PTP clock must synchronize only to a PTP port of its parent clock that is in the TIME_TRANSMITTER state. The synchronization must be received on, and only on, the port of the PTP clock that is in the TIME_RECEIVER state. However, in some instances it may be desirable for a PTP port to monitor a peer PTP port in a situation where the local PTP port is not in the TIME_RECEIVER state and/or the peer PTP port is not in the TIME_TRANSMITTER state.

The monitoring is enabled through the use of the portDS.monitorSender and portDS.monitorReceiver. A port with either portDS.monitorSender or portDS.monitorReceiver configured as TRUE will send and receive any messages required by its portDS.portState. When portDS.monitorSender is TRUE, and the portDS.portState is not TIME_TRANSMITTER, the PTP port also sends Announce, Sync, Follow_Up and Delay_Resp messages with the alternateTimeTransmitterFlag set to TRUE. When portDS.monitorReceiver is TRUE, and the portDS.portState is not TIME_RECEIVER, the PTP port also sends Delay_Req messages with alternateTimeTransmitterFlag set to FALSE. The message rates specified in clause 6.2.8 should be used for these specific messages sent when either portDS.monitorSender or portDS.monitorReceiver is TRUE.

These messages may allow a PTP port to acquire knowledge of the characteristics of the transmission path typically by estimating the Time Error on this port (i.e., the time difference between the local PTP clock using the TIME_RECEIVER port and the peer PTP clock connected via this port).

If the time error of the monitored port exceeds a threshold, this PTP clock may generate an alarm, which means that a fibre asymmetry, or a clock failure could be present in this PTP network. Note that this alarm is just used for PTP monitoring and should not trigger the Alternate BTCA. Additionally, the threshold used for this alarm should be properly configured to avoid inappropriate alarms.

Some examples of use cases where a PTP port may want to acquire knowledge of the characteristics of the transmission path are included in the next sub-clauses.

If the port detects that all expected ingress messages to compute the time error are not received, the PTSF-LOTM may be set as specified in clause 6.3.9.

The monitoring feature uses some of the functionality of alternate timeTransmitter (clause 17.4 of [IEEE 1588-2008], clause 17.3 of [IEEE 1588-2019]), limited to the ability to transmit and process PTP messages in the PTP port states as described above, with the alternateTimeTransmitterFlag set as appropriate.

Table G.1 captures various combinations of Port A and Port B states showing when Port A is able to monitor Port B. Some combinations may occur due to the use of timeTransmitterOnly, notTimeTransmitter, timeReceiverOnly or some other means to override the default behaviour.

Table G.1 – Combination of PTP port states with monitoring

Port A state	Port B state	Port A monitoring of port B is supported when	
		Port A configuration of portDS.monitorReceiver	Port B configuration of portDS.monitorSender
TIME_RECEIVER	TIME_TRANSMITTER	(Note 1)	(Note 2)
PASSIVE	TIME_TRANSMITTER	TRUE	(Note 2)
TIME_TRANSMITTER	TIME_RECEIVER	TRUE	TRUE
TIME_TRANSMITTER	PASSIVE	TRUE	TRUE
TIME_TRANSMITTER	TIME_TRANSMITTER	TRUE	(Note 2)
NOTE 1 – The configuration of portDS.monitorReceiver does not impact the operation of a PTP port in the TIME_RECEIVER state (i.e., it is not required to be TRUE). NOTE 2 – The configuration of portDS.monitorSender does not impact the operation on a PTP port in the TIME_TRANSMITTER state (i.e., it is not required to be TRUE).			

NOTE – Table G.1 lists the most typical combinations of PTP port states. The monitoring function described in this annex also applies to other combinations.

G.1 Use case 1 – PASSIVE port monitoring TIME_TRANSMITTER port

Figure G.1 shows an example illustrating where a PTP clock with a PTP port in the PASSIVE state (portDS.monitorReceiver is TRUE) monitors a peer PTP port in the TIME_TRANSMITTER state. As shown below, T-BC #4 has a PTP port in the PASSIVE state and it is desirable to monitor T-BC #2 that has a PTP port in the TIME_TRANSMITTER state. When the T-BC #4 PTP PASSIVE port receives the Sync and Delay_Resp message from the T-BC #2 PTP TIME_TRANSMITTER port, the T-BC #4 can obtain the time error (based on PTP timestamps) in reference to the local PTP clock (computed on the TIME_RECEIVER port, based on the time of T-BC #3 PTP TIME_TRANSMITTER port).

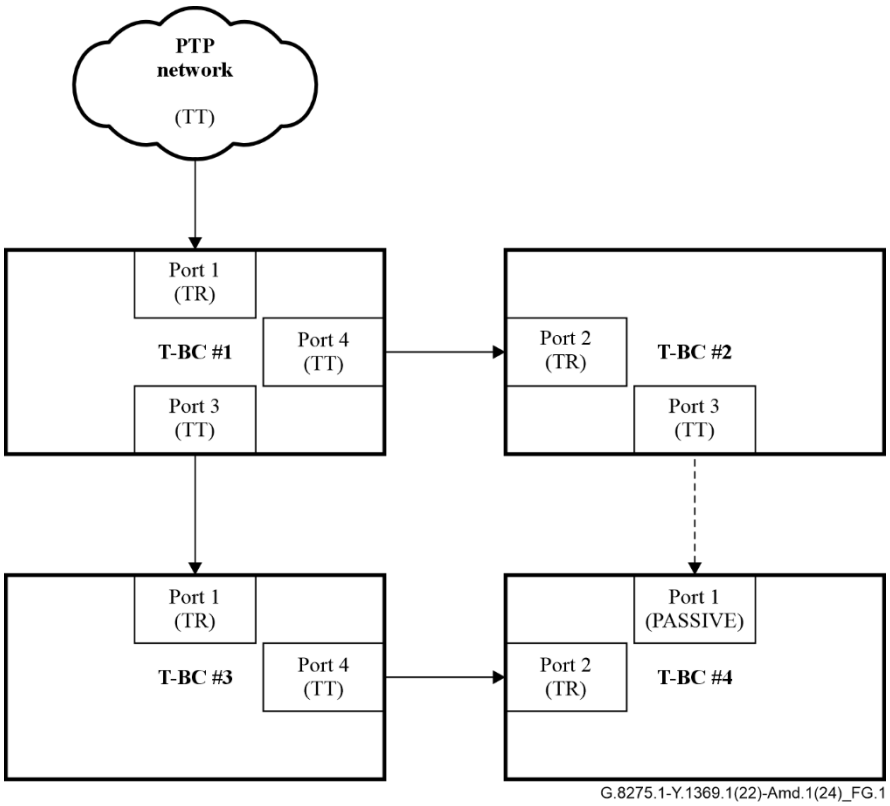


Figure G.1 – PASSIVE port monitors TIME_TRANSMITTER port

G.2 Use case 2 – TIME_TRANSMITTER port monitoring PASSIVE port

Figure G.2 shows an example illustrating where a PTP clock with a PTP port in the TIME_TRANSMITTER state (portDS.monitorReceiver is TRUE) monitors a peer PTP port in the PASSIVE state (portDS.monitorSender is TRUE). In the situation shown, Port 2 could be a PTP port or a virtual PTP port in the TIME_RECEIVER state preferring to synchronize to the local GNSS receiver. Port 1 in the TIME_TRANSMITTER state is used for monitoring of the PTP network, and through the monitoring function, the PTP clock is able to measure the time error on the PTP network: when the PTP TIME_TRANSMITTER port #1 receives the Sync and Delay_Resp messages from the peer clock's PASSIVE port, the PTP clock can estimate the time error of the TIME_TRANSMITTER port #1 in reference to the local PTP clock (computed on the TIME_RECEIVER port #2, based on the time of the PRTC).

Note that in Figure G.2 it is assumed that the PTP Clock Port 1 is in the TIME_TRANSMITTER state and the peer port in the PTP network is in the PASSIVE state, but that the opposite may also occur (where Port 1 is PASSIVE and the peer port in the PTP Network is TIME_TRANSMITTER, which is described in clause G.1) depending on the network planning and the Alternate BTCA.

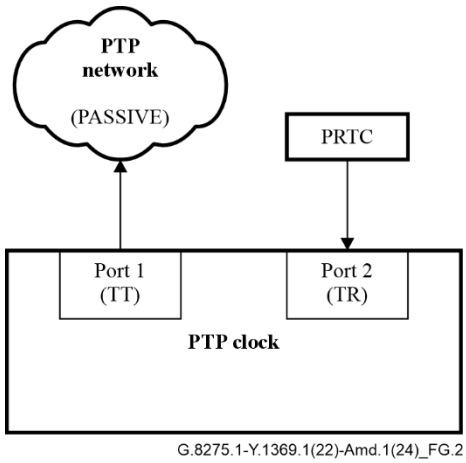


Figure G.2 – TIME_TRANSMITTER port monitors PASSIVE port

G.3 Use Case 3 – TIME_TRANSMITTER port monitoring TIME_TRANSMITTER port

Figure G.3 shows a use case where a PTP probe may monitor information from a PTP clock. Additionally, the PTP clock may likewise monitor information from the PTP probe. Considering a PTP probe working as a monitor function instead of a PTP reference, it is assumed that the PTP clock Port 2 is configured with portDS.timeTransmitterOnly TRUE in order to prevent the PTP probe from being selected by the Alternate BTCA of the PTP clock. To allow the PTP probe to monitor the PTP clock, the port of the PTP probe is configured with portDS.monitorReceiver TRUE. To allow the PTP clock to monitor the PTP probe, Port 2 of PTP clock is configured with portDS.monitorReceiver TRUE. Once both are configured as described there may be a full exchange of information between the PTP probe and the PTP clock. As a result, the PTP probe may full characterize/probe the PTP clock (receiving Sync and Delay_Resp), but also the PTP clock may gain the full knowledge of the PTP probe, by reception of Sync and Delay_Resp from the PTP probe: when the PTP TIME_TRANSMITTER port #2 receives the Sync and Delay_Resp messages from the probe's TIME_TRANSMITTER port, the PTP clock can estimate the time error of the TIME_TRANSMITTER port #2 in reference to the local PTP clock (computed on the TIME_RECEIVER port #1, based on the time of the network's TIME_TRANSMITTER port).

Note, for some use cases, it may be only necessary to monitor the timing of one direction but not both, e.g., a PTP probe monitors a PTP clock, or a PTP clock monitors a PTP probe to estimate its accuracy.

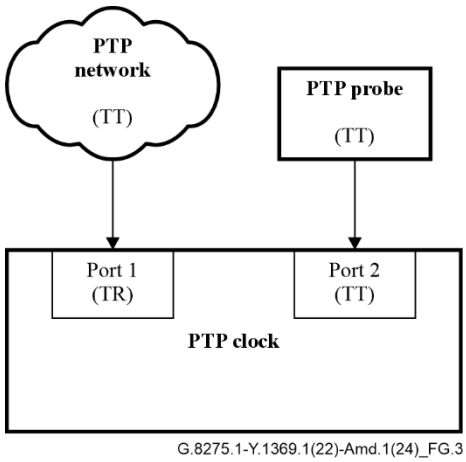


Figure G.3 – TIME_TRANSMITTER port monitors TIME_TRANSMITTER port

Annex H

Transport of PTP over OTN

(This annex forms an integral part of this Recommendation.)

The transport of PTP over OTN shall be performed by encapsulating the PTP messages into GFP-F frames as specified in clause 7.10 of [ITU-T G.7041], and then inserting the GFP-F frames into the OTUk OSMC as specified in clause 15.7.2.4 of [ITU-T G.709] and the FlexO OSMC as specified in clause 9.2.10 of [ITU-T G.709.1].

The PTP messages shall be timestamped as specified in clause 15.7.2.4.1 of [ITU-T G.709] and in clause 9.2.10.1 of [ITU-T G.709.1].

PTP messages may also be carried over the OSC OSMC as specified in clause 14.1 of [ITU-T G.709]. The PTP encapsulation in this latter case is vendor-specific.

All bits of the transportSpecific field (see clause 13.3.2.1 of (IEEE 1588-2008]) shall be transmitted as zero and shall be ignored by the receiver.

The value of the networkProtocol enumeration for the transport of PTP over OTN shall be F000 (hex).

Annex I

Transport of PTP over MTN

(This annex forms an integral part of this Recommendation.)

The transport of PTP over MTN is performed by encapsulating PTP messages into Ethernet frames that are transported by the MTN overhead as specified in clause 9.2.5 of [ITU-T G.8312]. The PTP event messages are timestamped. The timestamp generation is for further study.

The value of the networkProtocol enumeration for the transport of PTP over MTN is F001 (hex).

Annex J

Transport of PTP over FlexE links

(This annex forms an integral part of this Recommendation.)

The transport of PTP over FlexE shall be performed by encapsulating PTP messages into Ethernet frames that are transported by the FlexE overhead of the first FlexE Instance of the first PHY in the FlexE Group per clause 7.3.5 of [OIF FlexE IA]. The PTP event messages shall be timestamped as per clause 7.3.5 of [OIF FlexE IA].

The value of the networkProtocol enumeration for the transport of PTP over FlexE shall be F002 (hex).

Transport of PTP as a client of FlexE is for further study.

Annex K

Dynamic Monitoring based on PTSF (optional)

(This annex forms an integral part of this Recommendation.)

This annex is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

This annex describes an approach to dynamic monitoring of a peer PTP port by a PTP port with local PTSF. When this annex is implemented, a PTP port is additionally permitted to transmit Delay_Req messages and process received Delay_Resp messages in the following cases:

– either UNCALIBRATED, TIME_RECEIVER or PASSIVE state,

– or if portDS.SF is TRUE, regardless of the current port state,

– or if portDS.monitorReceiver is TRUE (as per Annex G of this Recommendation), regardless of the current port state.

The PTSF function is specified by clause 6.3.9.

When a PTSF event has occurred at a PTP port of a PTP clock, the PTP clock may set portDS.SF to TRUE according to clause 6.3.9. Upon the change of portDS.SF to TRUE, regardless of its current state, the PTP port is allowed to send a Delay_Req message and monitor the correct reception of Delay_Resp messages. This functionality allows the local PTP port to update the PTSF status. If it is verified that the PTSF has been cleared, portDS.SF of the PTP port is set to FALSE. As a result, the PTP port is considered as a candidate reference with normal handling of PTP messages and the Alternate BTCA can use Announce messages received on the port. If the WTR optional feature is used, the PTP port is considered as a candidate reference after the WTR timer expires.

The monitoring feature uses some of the functionality of the alternate timeTransmitter (clause 17.4 of [IEEE 1588-2008], clause 17.3 of [IEEE 1588-2019]), limited to the ability to transmit and process PTP messages in the PTP port states as described above, with the alternateTimeTransmitterFlag set as appropriate.

An example for the case of a T-BC is shown by Figure K.1.

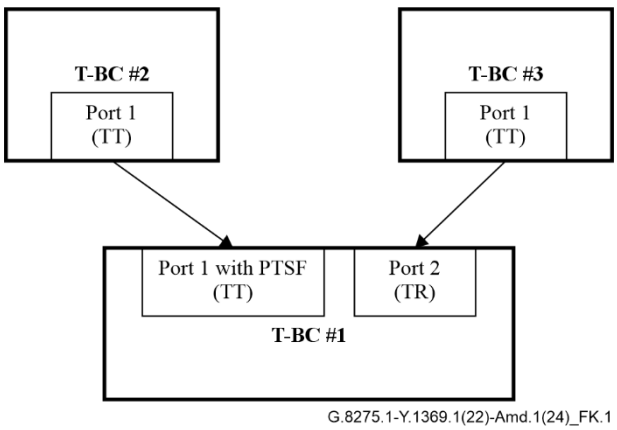


Figure K.1 – An example of PTSF restoration

In Figure K.1, Port 1 of T-BC #1 was previously in the TIME_RECEIVER state synchronizing with Port 1 of T-BC #2. After a while, T-BC #1 detects a PTSF event at Port 1 and changes its portDS.SF to TRUE. According to clause 6.3.9, T-BC #1 runs its Alternate BTCA, selects Port 1 of T-BC #3 as its new reference, and sets Port 2 to TIME_RECEIVER, Port 1 to TIME_TRANSMITTER (if configured with portDS.notTimeTransmitter as FALSE) or PASSIVE (if configured with portDS.notTimeTransmitter as TRUE).

In order to recover the PTSF failure of Port 1 of T-BC #1, that T-BC needs to continue to send messages towards T-BC#2 to detect when the SF condition is resolved. The event of portDS.SF being set to TRUE triggers Port 1 of T-BC #1 to continue to send Delay_Req messages. Port 1 of T-BC #2 in the TIME_TRANSMITTER state is able to reply with Delay_Resp messages. As a result, Port 1 of T-BC #1 will continue to receive Announce, Sync and Delay_Resp messages in order to determine if the signal fail condition has been cleared. If the failure is cleared, T-BC #1 sets portDS.SF of Port 1 to FALSE and then Announce messages received on Port 1 of T-BC #1 can be considered in the Alternate BTCA, which may select it as a TIME_RECEIVER port again. If the WTR optional feature is used, the PTP port is considered as a candidate reference after the WTR timer expires.

Annex L

Time/phase offset between local time reference and PTP (Optional)

(This annex forms an integral part of this Recommendation.)

This annex is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

L.1 Introduction

The time/phase offset between a local time reference (LTR) and a PTP reference can be used to detect faults in the network, as shown in Figure L.2.

A common use-case is to report the time difference between a wireless base station and a transport network as shown in Figure L.1.

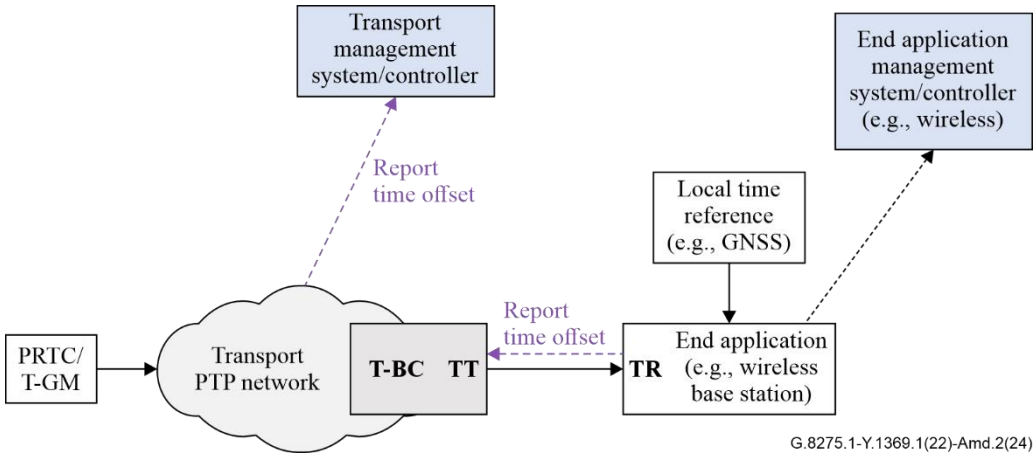


Figure L.1 – Example of a use-case for reporting LTR-PTP time/phase difference

In the use-case example shown in Figure L.1, if the end application management system has a direct connection to the transport management system, then it can report the time directly to the transport management system. However, in some cases, there is no direct connection between the end application management system and the transport management system, and in these cases, the LTR-PTP time/phase difference can be reported via the Signalling message with a TLV (see clauses L.2 and L.3) that is sent by the end application (e.g., a wireless base station) to the transport network.

As shown by Figure L.1, an end application (e.g., a base station) uses both LTR and PTP as its reference, one is active and the another is the backup. The time/phase offset between LTR and PTP can be conveyed in a PTP Signalling message with a TLV from the end application to the T-BC, and a transport management system may use this information to detect a fault in the network.

Clauses L.2 and L.3 specify the message format of the Signalling message and its TLV. Note, the signalling message mapping and encapsulation shall follow clauses 6.2.6 and 6.2.7.

NOTE 1 – This annex covers the use case of a T-TSC integrated in the end application. The use case of a T-BC integrated in the end application is for further study.

NOTE 2 – This annex covers the use case of a direct link between the T-BC and the end application, so that each TT port of the T-BC only receives one reported time offset from the end application.

L.2 Signalling message for time difference between LTR and PTP references

Table L.1 lists the Signalling message fields.

Table L.1 – Signalling message for time/phase offset between LTR and PTP references

Signalling message fields	Octets
header	34
targetPortIdentity	10
TLV	54

1) **header**: it shall follow the definition of clause 13.3 of [IEEE 1588];
2) **targetPortIdentity**: Its data type is PortIdentity, and it should be the sourcePortIdentity of the received Announce message at a PTP port sending the Signalling message. If unknown, it shall be set to all ones.
3) TLV: see clause L.2.

L.3 TLV for Time/phase offset between LTR and PTP references

Table L.2 defines the TLV message fields.

Table L.2 – TLV for Time/phase offset between LTR and PTP references

TLV fields		Octets
tlvType		2
lengthField		2
organizationId		3
organizationSubType		3
dataField	clockMode	1
	flags	1
	ptpGrandmasterID	8
	offsetFromPtpToLTR	8
	messageInterval	2
	Reserved	24

1) **tlvType**: 0x0003 (ORGANIZATION_EXTENSION)
2) **lengthField**: 50
3) **organizationId**: 00-19-A7 (ITU-T OUI code)
4) **organizationSubType**: 00-00-04
5) **clockMode**: as specified in clause 6.2.9
6) **flags**
   a) bit 0: **withLTR**, TRUE if the end application has an LTR reference, otherwise FALSE, if the end application does not have an LTR.
   b) bit 1: **LTRValid**, TRUE if the LTR status of the end application is valid, otherwise FALSE, if the LTR status is invalid. This status is implementation specific.

c) bit 2: **timeStatus**, TRUE if the time of the end application is normal, e.g., there is no clock alarms on the end application, otherwise FALSE, if the time of the end application is unnormal, e.g., the time offset calculated from the active reference is not stable. This status is implementation specific.

d) bit 3: **ptpReference**, TRUE if the active time reference of the end application is PTP, otherwise FALSE, if the active time reference is LTR.

e) bit 4 ~ 7: reserved.

7) **ptpGrandmasterID**: the grandmaster identifier (ID) of the PTP Announce message received at the end application.

8) **offsetFromPtpToLTR**: the mean time offset of the LTR reference and the PTP reference (LTR time minus PTP time) within the Signalling message interval, and its type is TimeInterval (see clause 5.3.2 of [IEEE 1588-2019]), whose unit is $1/2^{16}$ ns.

9) **messageInterval**: the interval of the Signalling message sent by the end application, and the data type is UInteger16. Its unit is seconds, and the default value is 60.

10) **Reserved**: 24 octets, used for some future extension.

NOTE – Additional details related to the transmission and reception of Signalling messages for this TLV are for further study.

Annex M

Synchronization service availability performance metrics (Optional)

(This annex forms an integral part of this Recommendation.)

This annex is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

Annex G of [ITU-T G.8275] specifies the general concept of synchronization service availability performance metrics. Particular criteria for full timing support networks are specified in this annex.

Two independent criteria are considered for the synchronization service availability performance metrics in such a profile: one for the frequency layer and one for PTP. For the frequency layer, the criterion is for further study. For PTP, the criterion to determine either degraded 1 second (D1S) vs nominal 1 second (N1S) is parentDS.grandmasterClockQuality.clockClass, using a clockclass threshold: Cth.

Cth is configured using the defaultDS.clockclassAvailabilityThreshold dataset member. Its format is the same as that of the parentDS.grandmasterClockQuality.clockClass dataset member, and the notion of hierarchy between clockclass values, (i.e., which value is better than or equal to Cth) is per [IEEE-1588], i.e., 'the smaller the better'. The default value of Cth is 6.

As an example, if Cth is 6, then N1S is determined when the PTP clockclass value is 6 or better, during the entire period 1 second, while D1S is determined when the PTP clockclass value has been strictly worse than 6 any time during the period of 1 second.

ptpServiceAvailabilityPerformanceMetricsDS is used to report the available seconds (AS) and unavailable seconds (UAS) metrics of the PTP service. See the detailed description in clause G.3 of [ITU-T G.8275].

Other availability criteria may be considered; this is for further study.

Annex N

Enhanced accuracy metrics TLV option (Optional)

(This annex forms an integral part of this Recommendation.)

This annex is optional but, if implemented, it is necessary for the equipment to conform to the requirements contained herein.

N.1 Introduction

This annex is based on [IEEE 1588a-2023] and modified to optimize the use of the enhanced accuracy metrics TLV for a telecom network.

PTP networks consisting of grandmaster clocks, boundary clocks, transparent clocks and ordinary clocks can more accurately communicate the PTP timing accuracy by using the enhanced accuracy metrics TLV option. The grandmaster time metrics are communicated in Announce messages using the enhanced accuracy metrics TLV option allowing GM time information to be shared with the end user equipment compared to only relying on the GM clockAccuracy attribute. It also allows the end user equipment to get information on the time accuracy of the PTP path that the PTP messages have traversed (i.e., through T-BCs and T-TCs).

Additionally, the enhanced accuracy metrics TLV can be used to indicate the impact of the interworking function (IWF) node in network performance, as in some deployments, it is possible to have a mix of timing trails to the timeReceiver, for example:

- One timing trail from the primary reference time clock (PRTC) via an IWF;
- Another timing trail from the PRTC via a pure full timing support (FTS) network.

In such cases, it may be useful for the IWF node to indicate the impact of its upstream timing trail, so that downstream PTP clocks can choose the most suitable timing trail for the end application. The values to be indicated in the enhanced accuracy metrics TLV are implementation-specific based on details of the IWF node and its use in the network.

Two use cases of the enhanced accuracy metrics TLV are given in the following clause.

N.2 Use cases

N.2.1 Network deploying T-BCs with different clock classes

Figure N.1 shows a use-case of a network deploying T-BCs with different clock classes. If complying with the Best TimeTransmitter Clock Algorithm (BTCA) defined in this Recommendation that does not support the enhanced accuracy metrics TLV, T-BC3 will select a synchronization path composed of T-BC class B nodes based on the stepsRemoved comparison, resulting in a selection of a suboptimal synchronization path. A more optimal synchronization path might be possible by using an alternate BTCA with information from the enhanced accuracy metrics TLV taken into consideration.

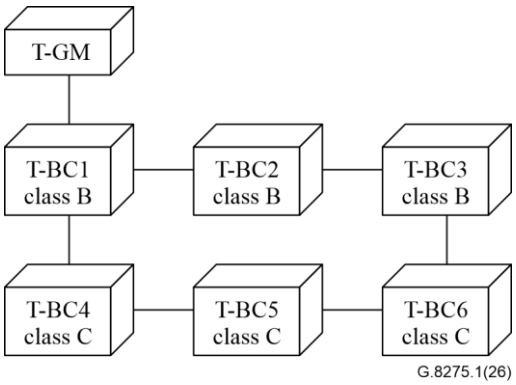


Figure N.1 – Network deploying different clock classes

N.2.2 Network deploying a mix of T-TCs and T-BCs

Figure P.2.1 in [IEEE 1588a-2023] shows a network architecture incorporating both transparent clocks and boundary clocks, which can be adapted for telecom applications using T-TCs and T-BCs. Due to T-TCs not updating path information (e.g., stepsRemoved), Port 1 of T-BC3 will be a timeReceiver port based on BTCA defined in this Recommendation that does not support the enhanced accuracy metrics TLV, even though the path to the telecom grandmaster (T-GM) involves a greater number of T-TCs. This could result in a selection of a suboptimal synchronization path. A more optimal synchronization path might be possible by using an alternate BTCA with information from the enhanced accuracy metrics TLV taken into consideration.

N.3 High-level design requirements

To gain the full benefit of the enhanced accuracy metrics TLV option, all PTP clocks in the PTP domain shall support the option. In network scenarios, where only part of the PTP clocks in the PTP domain supports this option, T-GMs should support this option.

NOTE 1 – Only T-GM is allowed to originate the enhanced accuracy metrics TLV.

NOTE 2 – The detailed technical requirements related to T-TC are for further study.

NOTE 3 – The use of the enhanced accuracy metrics TLV in an alternate BTCA is for further study.

N.4 Technical specification

N.4.1 ENHANCED_ACCURACY_METRICS TLV

ENHANCED_ACCURACY_METRICS TLV, defined by [IEEE 1588a-2023], with the format adapted for this profile is specified in Table N.1.

Table N.1 – ENHANCED_ACCURACY_METRICS TLV for telecom profile

Bits								Octets	TLV offset
7	**6**	**5**	**4**	**3**	**2**	**1**	**0**		
tlvType (Note 1)								2	0
lengthField (Note 2)								2	2
bcHopCount (Note 3)								1	4
tcHopCount (Note 4)								1	5
exclusionFlags (Boolean[0]–Boolean[7]) (Note 5)								1	6
EVSII	EMSII	EVDI	EMDI	EVTI	EMTI	EVGI	EMGI		
exclusionFlags (Boolean[8]–Boolean[16]) (Note 6)								1	7
Reserved						EVSMI	EMSMI		
maxGmInaccuracy (Note 7)								8	8
varGmInaccuracy (Note 7)								8	16
varDynamicInaccuracy (Note 8)								8	24
maxStaticInstanceInaccuracy (Note 8)								8	32
maxStaticMediumInaccuracy (Note 8)								8	40
varStaticMediumInaccuracy (Note 8)								8	48

NOTE 1 – Set by T-GM to 0x4001.
NOTE 2 – Set by T-GM to 52.
NOTE 3 – Initiated by T-GM to 0 updated by T-BC.
NOTE 4 – Initiated by T-GM to 0 updated by T-TC.
NOTE 5 – Initiated by T-GM to binary 1001 1100.
NOTE 6 – Initiated by T-GM to binary 0000 0000.
NOTE 7 – Set by T-GM.
NOTE 8 – Initiated by T-GM to 0, updated by T-BC/T-TC.

N.4.2 Timing path accuracy

ENHANCED_ACCURACY_METRICS TLV is updated along the path by each T-BC/T-TC supporting this TLV. The accumulation shall be via addition of the PTP instance's own contribution to the received values of each applicable metric X:

$$\text{transmitted}(X) = \text{received}(X) + \text{local}(X)$$

where X is maxStaticInstanceInaccuracy or varDynamicInaccuracy.

The accumulation shall be via addition of the medium/link contribution (providing timing to the PTP instance) to the received values by the PTP instance for each applicable metric Y:

$$\text{transmitted}(Y) = \text{received}(Y) + \text{local}(Y)$$

where Y is maxStaticMediumInaccuracy or varStaticMediumInaccuracy and local refers to medium/link providing timing.

Table N.2 – Path time error accumulation

maxGmInaccuracy	$\max\|TE\|_{T-GM}$	cTE_{T-GM}
varGmInaccuracy	0	dTE_{T-GM}^2
maxStaticInstanceInaccuracy	$\sum_{i=1}^{N} cTE_{T-BC/T-TC_i}$	
varDynamicInaccuracy	$\sum_{i=1}^{N} dTE_{T-BC/T-TC_i}{}^2$ (Note 1)	
maxStaticMediumInaccuracy	0 if unknown, else $\sum_{i=1}^{N+1} cTE_{medium/link_i}$ (Note 2)	
varStaticMediumInaccuracy	0 if unknown, else $\sum_{i=1}^{N+1} dTE_{medium/link_i}{}^2$ (Note 1, 2)	
NOTE 1 – $dTE_i{}^2$ value is for further study. NOTE 2 – Number of media/links between T-GM and PTP instance updating metrics is N+1, where N is number of T-BC/T-TC in between.		

For the nodes, the formula for calculating the path contribution is according to this formula:

$$\text{path}_{TE} = \text{maxStaticInstanceInaccuracy}_{\text{received}} + \text{maxStaticMediumInaccuracy}_{\text{received}} + \text{maxStaticMediumInaccuracy}_{\text{link}} + \sqrt{[\ \text{varDynamicInaccuracy}_{\text{received}} + \text{varStaticMediumInaccuracy}_{\text{received}} + \text{varStaticMediumInaccuracy}_{\text{link}}]}$$

N.4.3 Deployment considerations

The following deployment scenarios describe various situations that can occur and shall be considered by implementers and network designers:

1) All clocks support the enhanced accuracy metrics TLV, and have this option enabled;
2) The T-GM supports the enhanced accuracy metrics TLV, have this option enabled, and T-BC/T-TC supports the option, but it is disabled;
3) The T-GM supports the enhanced accuracy metrics TLV, and have this option enabled, and T-BC/T-TC does not support this option.

The detailed deployment considerations for these scenarios are for further study.

The enhanced accuracy metrics TLV option can be very useful for an end application to obtain an estimate of the time error for the PTP path. This knowledge can be used by the end application to estimate better how long it can be in holdover in case of loss of synchronization. It can also help to optimize certain functions by having a good estimate of the current time error.

Details are for further study.

Appendix I

Considerations on the use of transparent clock

(This appendix does not form an integral part of this Recommendation.)

The integration of the transparent clock in this profile is considered to be of particular interest for applications such as 2-port type of devices.

For more complex topologies and multiport devices some careful analysis is recommended. In particular, some specific node configuration may be required if the operator wants to prevent multicast packets from flooding the network.

Some options could be considered, e.g., by configuring the T-TCs with insertion of an [IEEE 802.1Q] VLAN tag in the frames carrying PTP. In this case all T-TCs in the PTP communication path where this option is used should support this option. The last T-TC node would need to remove the [IEEE 802.1Q] VLAN tag. By using this option to connect the physical ports of the T-GM, T-BC, and T-TSCs through [IEEE 802.1Q] VLAN virtual connections across the T-TCs, multicast flooding can be avoided. In this way the PTP port of a T-GM, T-BC or T-TSC would never process a [IEEE 802.1Q] VLAN tagged frame.

The related operational implications would need to be carefully considered.

NOTE – In case of multiport T-TCs with FlexE interfaces, it might be required to carry multiple PTP messages on the overhead of the outgoing FlexE interfaces. The related overhead bandwidth limitations need to be taken into account.

Appendix II

Considerations on the transmission of Delay_Req messages

(This appendix does not form an integral part of this Recommendation.)

This appendix discusses the requirements defined in [IEEE 1588] for the transmission of *Delay_Req* messages when using the default uniform distribution defined in bullet 3 of clause 9.5.11.2 of [IEEE 1588-2008] and bullet (1) of clause 9.5.11.2 of [IEEE 1588-2019]. This uniform distribution is not used in the PTP telecom profile defined in this Recommendation; a profile-specific distribution has been defined in clause 6.2.8.

The second dashed item of the requirements defined in clause 9.5.11.2 of [IEEE 1588-2008] and bullet (b) of clause 9.5.11.2 of [IEEE 1588-2019] controls the variability of the times between successive *Delay_Req* messages. It is analogous to the corresponding requirement for the sending of *Sync* and *Announce* messages, given in clause 7.7.2.1 of [IEEE 1588]. However, a key difference is that, while the requirement for *Sync* messages applies to the population of inter-message intervals, the requirement for *Delay_Req* messages applies only to the mean of the population.

To be more precise, assume that a population of N inter-message intervals has been measured, and let T_j be the measured values, $j = 1, 2, \ldots, N$. The sample mean, m, is just the numerical average, i.e.:

$$m = \frac{1}{N}\sum_{j=1}^{N} T_j \tag{II.1}$$

Let T_{min} be the minimum *Delay Request* interval; it is equal to $2^{\text{portDS.logMinDelayReqInterval}}$ s. The second dashed item above states that the mean of the distribution must be greater than or equal to T_{min} with 90% or greater statistical confidence.

The statistical test for this is well-known and is based on the fact that the distribution of m approaches a normal distribution as N becomes large (i.e., it is based on the central limit theorem). Let s be the standard deviation of the distribution of the inter-message times, i.e., the distribution of the T_j. Let $z_{0.90}$ be the 90th percentile of the standard normal distribution; it is given by $z_{0.90} = 1.281$. Then, the probability that the true mean of the distribution exceeds the quantity:

$$q_{0.1} = m - z_{0.90}\sqrt{\frac{\sigma}{N}} \tag{II.2}$$

is 0.9, i.e., 90%. The probability that the mean of the distribution is less than this value is 0.1. In addition, if σ is not known, the sample standard deviation, s, may be used in equation (II.2) and the Normal distribution is replaced by the Student-t distribution with $N - 1$ degrees of freedom. The sample standard deviation is given by:

$$s = \left[\frac{1}{N-1}\sum_{j=1}^{N}(T_j - m)^2\right]^{1/2} \tag{II.3}$$

To meet the requirement, the quantity $q_{0.1}$ must exceed T_{min}. It is seen from equation (II.2) that $q_{0.1}$ approaches m as N approaches infinity. Since m converges to the mean of the distribution of inter-message times as N approaches infinity, the requirement can be met for sufficiently large N as long as the mean of the distribution exceeds T_{min}. The mean of the distribution must exceed T_{min}; the requirement cannot be met if the mean of the distribution is exactly equal to T_{min} or less than T_{min}.

If it is chosen to comply with the requirement of the third dashed item of clause 9.5.11.2 of [IEEE 1588-2008] or bullet (1) of clause 9.5.11.2 of [IEEE 1588-2019], one way of meeting this requirement is to increase the upper end of the probability distribution by 10%. If this is done, the transmission times are selected such that the interval between successive *Delay_Req* messages is taken from a uniform distribution over the interval between 0 and $2.2T_{min}$. A new random value for the transmission interval is computed for each message transmitted. When computing the mean *Delay_Req* interval from measured samples to check if it exceeds T_{min} with 90% or greater statistical confidence (if the third dashed item is used), the number of measured samples N must be at least 1000. The granularity of the distribution must be less than or equal to 1/16 of the *Sync* interval.

If it is chosen to meet the requirement of the fourth dashed item of clause 9.5.11.2 of [IEEE 1588] or bullet (2) of clause 9.5.11.2 of [IEEE 1588-2019], a *Delay_Req* message is transmitted as soon as possible after receipt of a *Sync* message, subject to not violating the second dashed item.

Appendix III

Considerations on the choice of the PTP Ethernet multicast destination address

(This appendix does not form an integral part of this Recommendation.)

This PTP profile supports both the non-forwardable multicast address 01-80-C2-00-00-0E and forwardable multicast address 01-1B-19-00-00-00 when the PTP mapping defined in Annex F of [IEEE 1588-2008] or Annex E of [IEEE 1588-2019] is used.

The Ethernet multicast address to be used depends on the operator policy; further considerations are provided hereafter.

Layer 2 bridging function associated with the PTP port of a T-BC or T-TC should not forward any frame with destination MAC address 01-1B-19-00-00-00; this could be done by properly provisioning this multicast address in the filtering database.

Option 1 – Use of the non-forwardable multicast address 01-80-C2-00-00-0E

Some network operators object to PTP messages being forwarded through PTP-unaware network equipment.

The use of the non-forwardable multicast address 01-80-C2-00-00-0E guarantees this property most of the time (exceptions exist for some older Ethernet equipment).

Therefore, in the case of network equipment misconfiguration (e.g., if the PTP functions are not enabled in PTP-aware network equipment), the use of this multicast address prevents incorrect distribution of synchronization, since the PTP messages will be blocked by the PTP-unaware network equipment.

Option 2 – Use of the forwardable multicast address 01-1B-19-00-00-00

Some network operators consider that using a forwardable multicast address is more flexible and that it is preferable to forward the PTP messages to keep the synchronization link running in case some equipment is misconfigured as non PTP nodes, although there are potentially risks of performance degradation. The network management system (NMS) will easily find the misconfiguration and will send alarms.

However, it is possible to block the PTP messages by properly provisioning this multicast address in the filtering database of each Ethernet equipment.

Appendix IV

Considerations on the use of priority2

(This appendix does not form an integral part of this Recommendation.)

The PTP attribute priority2 is configurable in this profile. In some special circumstances, the use of the priority2 attribute can simplify network management. This appendix describes two use cases; other possible cases are for further study.

NOTE – Additional considerations on the use of priority2 are provided in Appendix XVI of this Recommendation.

Case 1

Operators can configure the PTP attribute priority2 to make all of the T-BCs either traceable to one T-GM, or traceable to two different T-GMs at the same time.

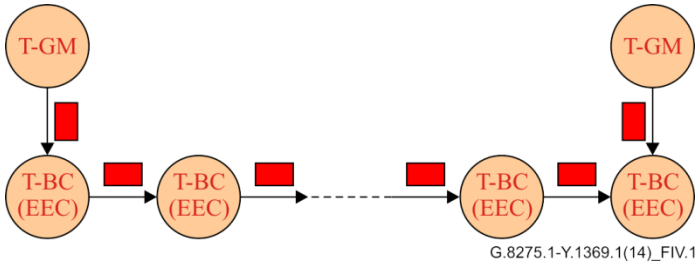


Figure IV.1 – Use of priority2 with two T-GMs in the network

For example, in Figure IV.1, if all other PTP attributes of the two T-GMs are the same, and the two T-GMs are configured with the same priority2 value, each T-BC will select the T-GM with the shortest path. If the two T-GMs are configured with different priority2 values, all of the T-BCs will synchronize to the T-GM with the smallest priority2 value.

Case 2

Operators can configure the PTP attribute priority2 to prevent the T-BCs of an upstream network from synchronizing with the T-BCs of a downstream network when the T-GM is in failure.

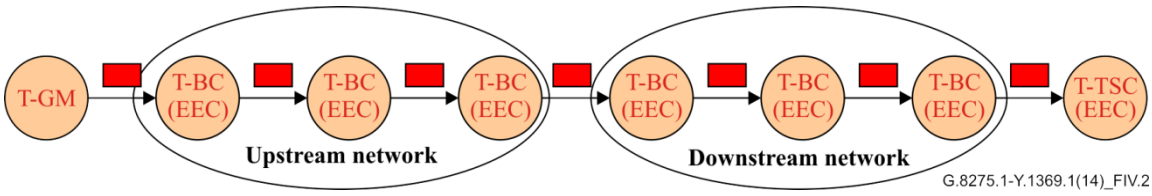


Figure IV.2 – Use of priority2 with T-BCs of different network layers

For example, in Figure IV.2, if all other PTP attributes of all of the T-BCs are the same, and the PTP attribute priority2 of all of the T-BCs is configured with the same value, then when the T-GM is in failure, the T-BCs in the upstream network can synchronize with the T-BCs in the downstream network, depending on the clockIdentity values of all of the T-BCs. If the T-BCs in the upstream network are configured with a smaller priority2 value than the T-BCs in the downstream network then, when the T-GM is in failure, the T-BCs in the downstream network will synchronize to the T-BCs in the upstream network.

Appendix V

Description of PTP clock modes and associated contents of Announce messages

(This appendix does not form an integral part of this Recommendation.)

Description of PTP clock modes and associated contents of Announce messages are described in Appendix VIII of [ITU-T G.8275].

Appendix VI

Operations over link aggregation

(This appendix does not form an integral part of this Recommendation.)

When two devices embedding PTP clocks compliant with this profile are connected via a link aggregation (LAG) as defined in [b-IEEE 802.1AX], each physical link should be accessed directly to transmit PTP messages, bypassing the LAG. This method prevents potential asymmetries that may be present when the forward and reverse paths are delivered over different links belonging to the LAG.

Alternative solutions taking advantage of some link aggregation (LAG) features exist, such as the bidirectional congruity as defined by [b-IEEE 802.1AX]. For the scenarios currently considered, the insertion of a VLAN tag in the frames carrying PTP messages is not allowed. However, the bidirectional congruity alternative solution can be applied to untagged PTP frames by assigning a conversation ID equal to zero to a given physical link. These solutions are for further study.

Appendix VII

Relationship between clockClass and holdover specification

(This appendix does not form an integral part of this Recommendation.)

The clockClass values that are used in this profile are described in Table 6. The values may be divided into four different categories:

1) T-GM locked to a PRTC or ePRTC;
2) T-GM or T-BC in holdover, within holdover specification;
3) T-GM or T-BC in holdover, outside of holdover specification;
4) TimeReceiver clock, or clock that has not been synchronized.

A short footnote (Note 1 of clause 6.4) refers to Appendix V of [ITU-T G.8271.1] for more information on the meaning of "in holdover, within holdover specification" and "in holdover, out of holdover specification". That appendix describes possible budget models for the synchronization performance. The exact model depends on the operator's network and design parameters, but each budget is broken down into several components:

1) PRTC/T-GM allocation;
2) Random time error from noise accumulation through the network (dTE);
3) Node asymmetry (cTE, the sum of the asymmetry of all the nodes in the system);
4) Link asymmetry (cTE, the sum of the asymmetry of all the links in the system);
5) Holdover budget;
6) End application budget.

Table V.1 of [ITU-T G.8271.1] shows that for one example budget, 400 ns may be allocated to holdover within the network (this is termed failure scenario (b) in the table). The operator may have different budget allocations, depending on their deployment scenario. The 400 ns holdover budget is allocated to the entire synchronization chain, and not to an individual clock.

The intended operation of the T-GM is, therefore, as follows:

– When the T-GM is synchronised to a PRTC locked to GNSS, it outputs clockClass 6.
– If the PRTC loses its connection to GNSS, it enters holdover. The T-GM should degrade the advertised clockClass to indicate "in holdover, but within holdover specification" (clockClass 7).
– The T-GM estimates when the holdover budget will potentially be exceeded. Factors to consider include the known quality of any external frequency support (e.g., synchronous Ethernet (SyncE) quality level (QL)), temperature variations, and/or the quality of the internal oscillator.
– When the T-GM considers that the clock is out of holdover specification (i.e., it is now estimated to have drifted by more than the holdover budget), the T-GM will advertise a clockClass of 140, 150 or 160.

In the event of a network failure, where the T-GM is disconnected from the synchronization chain, a T-BC will take over as the grandmaster of the chain. That T-BC will be operating in holdover. The clockClass that the T-BC is allowed to advertise depends on the clockClass of the T-GM to which it was synchronized prior to losing connectivity.

For example, if the T-BC was synchronized to a T-GM of clockClass 6, none of the holdover budget will have been consumed, and therefore the T-BC may use a clockClass indicating "within holdover specification" (e.g., clockClass 135). This value is chosen to be lower (i.e., better) than that of a T-GM that is out of holdover specification, since the T-BC is likely to have more accurate time because it

has been locked to a traceable time source more recently. Therefore, if the two clocks (a T-BC in holdover, within holdover specification and a T-GM in holdover, but out of holdover specification) are compared in the Alternate BTCA operation of a subsequent clock, the subsequent clock will synchronize to the T-BC that is within holdover specification instead of a T-GM that is out of holdover specification.

In another example, if the T-BC was synchronized to a T-GM indicating that the T-GM is in holdover but out of holdover specification (e.g., parentDS clockClass 140, 150 or 160), the T-BC should also use a clockClass indicating "out of holdover specification" (e.g., defaultDS clockClass 165). This is because the T-GM was indicating that, in its estimation, the holdover budget has already been consumed.

In another example, if the T-BC was synchronized to another T-BC, and the other T-BC indicates that it is in holdover but out of holdover specification (e.g., parentDS clockClass 165), the original T-BC should also use a clockClass indicating "out of holdover specification" (e.g., defaultDS clockClass 165). This is because the other T-BC was indicating that, in its estimation, the holdover budget has already been consumed.

In a final example, if the T-BC was synchronized to a T-GM or another T-BC that was already in holdover but still within the holdover specification, the original T-BC could indicate "within holdover specification". However, some of the holdover budget will have already been consumed by the T-GM or T-BC. If it is not known how much of the budget is left, the T-BC should indicate "out of holdover specification".

NOTE 1 – When the term clockClass is used with respect to the property of the individual PTP clock (T-GM, T-BC, T-TSC) it is referring to dataset member defaultDS.clockQuality.clockClass.

NOTE 2 – If a T-BC was synchronized to a T-GM or to another T-BC that was in free-run mode (advertising clockClass 248), the T-BC should continue to advertise clockClass 248 even after it loses its reference.

Appendix VIII

Considerations on a T-TSC connected to an end application

(This appendix does not form an integral part of this Recommendation.)

The default T-TSC clockclass 255 implies that the T-TSC will always lock to an external PTP reference when available.

The actual synchronization source ultimately used by the end application depends on the applicable synchronization needs. This process is out of the scope of this Recommendation.

As an example, the decision to use the PTP reference that has been selected by the T-TSC (e.g., instead of entering holdover), could depend on the actual clockQuality, frequencyTraceable flag, timeTraceable flag, and synchronizationUncertain flag associated with the external PTP reference. Additional aspects related to performance monitoring of the external reference might also be considered. This is implementation specific.

As an example, when it is required to meet the network timing requirements as per e.g., [ITU-T G.8271.1], it would be necessary for the external PTP reference to have clockClass 6, 7 or 135 and for the timeTraceable flag is TRUE in order to be used by the end application. When this condition is not met, the end application may decide to enter holdover (either using the internal oscillator or being driven by SyncE).

NOTE – The specific behaviour for a T-TSC embedded in the end application is outside the scope of this Recommendation. It is assumed that interoperability with the profile is maintained.

Appendix IX

Calculation of offsetScaledLogVariance for T-GM timed by PRTC or ePRTC

(This appendix does not form an integral part of this Recommendation.)

Calculation of offsetScaledLogVariance for T-GM timed by PRTC or ePRTC is described in Appendix XIV of [ITU-T G.8275].

Appendix X

Description of a T-BC extended clockClass application

(This appendix does not form an integral part of this Recommendation.)

Table 6 of [ITU-T G.8275.1] defines a T-BC extended clockClass application. This appendix describes the purpose of this extended application, and its corresponding value amendments, including defaultDS dataset member specifications of Annex A of [ITU-T G.8275.1] (corresponding to Table A.1) and T-BC Announce message contents of Appendix V of [ITU-T G.8275.1] (corresponding to Table V.3).

X.1 Purpose of T-BC extended clockClass application

PTP is the only synchronization source in an [IEEE 1588] network, therefore the clockClass values have only been applied to the PTP clock for indicating its inherent characteristics. The clockClass in Table 5 of [IEEE 1588-2008] and Table 4 of [IEEE 1588-2019], such as 6,7,52,187, only represents the T-GM. The clockClass may not reflect the change of synchronization states in the T-BCs deployed in the network.

Since it is possible that there are two separate synchronization sources (GNSS time and physical frequency) in a telecom network, an extension mechanism of the clockClass range to cover the case of both T-GM and T-BC changes is provided. The definition of the clockClass values is extended into two parts:

1) for a T-GM, the range of clockClass includes 6, 7, 140, 150 and 160. The clockClass values of 140/150/160 represent the time output of the T-GM when it is in time holdover instead of tracking to GNSS. The time quality is consistent with the clock quality of the frequency category 1, 2, and 3 respectively;
2) for a T-BC, the clockClass includes 135 and 165. The value of 135 means that when a T-BC is in time holdover, the output time signal is within specification, clockClass 165 means that the output time signal is out of time holdover specification.

The case of two separate sources has been considered in [ITU-T G.8271.1] HRM-2 and HRM-3. In order to extend the advantages of two separate sources, it is better to treat a T-BC differently when it is traceable to a frequency source of a different quality. The clockClass of 165 could be assigned to a T-BC when the T-BC is traceable to a frequency in QL-SEC, and the T-BC could share the clockClass value of 140/150/160 with the T-GM when it is traceable to a different quality of frequency. In this way, the appropriate T-BC can be selected to be the GM of the network so that better time signals can be sent to the end-users.

For a telecom network, it is crucial to make sure that the best clock selection principle is valid in any PTP network, especially in the case where there exists an isolated time region (which cannot receive any signals from the T-GM) in the network. In this region, if the T-BC has a preferred frequency quality, the T-BC can act as a T-GM and send out precise time/phase signals (where the signal quality is still close to UTC). In telecom networks, from the perspective of synchronization performance, any of the T-BCs could supersede the old T-GM to be the new PTP source of the network.

X.2 defaultDS dataset member specifications under this extended application

In some cases, the values of clockClass/accuracy/variance will likely be the same as that delivered by the announce packets from the T-GM and T-BC per Table 6/Note 3. As such, the T-BC should synchronize the T-GM. It is suitable for priority 2 to achieve this goal, and the range of priority 2 should be divided into two parts, 0-127 for the T-GM, 128-255 for the T-BC.

In this PTP profile, the clock attribute priority2 is configurable.

If this feature is used, then the *priority2* for the T-GM should be set to a lower value compared to T-BCs, in particular the range for *priority2* could be allocated as follows:

– for T-GM, the range {0-127};

– for T-BC, the range {128-255}.

As an example of a default value for *priority2*, it is suggested to set the T-GM to 100 and the T-BC to 128.

Appendix XI

Considerations on native access equipment

(This appendix does not form an integral part of this Recommendation.)

In some deployments, native access equipment such as x-type digital subscriber line (xDSL), x-type passive optical network (xPON) and microwave equipment may have implemented T-BC functionality. This appendix provides information on how such equipment may be modelled as a pair of T-BCs defined in this Recommendation. The following Figure XI.1 shows an example of equipment that is transferring the PTP timing between native PTP and native access media. Examples of such equipment may be an xPON optical line terminal (OLT) and ONU, an xDSL digital subscriber line access multiplexer (DSLAM) and remote terminal digital subscriber line access multiplexer (RT-DSLAM), or microwave equipment. Only one PTP port and one Special precision time protocol (PTP) port are shown in the diagram, although the equipment may contain multiple ports. Within this simplified diagram there are two PTP ports on the T-BC; one is a normal PTP port and one is a Special precision time protocol (PTP) port. On both the PTP port and the Special PTP port, the Announce information and Signalling messages are handled according to normal T-BC operation. The difference is that on the PTP port, the timing messages (Sync, Delay_Req, Delay_Resp) are handled normally, while on the Special PTP port they are not present as they are replaced by native timing messages. If the timing flow hierarchy is known in advance (such as that of a PON OLT, which will only transfer timing downstream on its Special PTP ports) then the PTP ports, Special PTP ports, or the PTP clock may be configured with timeTransmitterOnly or timeReceiverOnly parameters (such as a PON OLT configured with Special PTP ports as timeTransmitterOnly TRUE, or a PON optical network unit (ONU) configured with timeReceiverOnly TRUE).

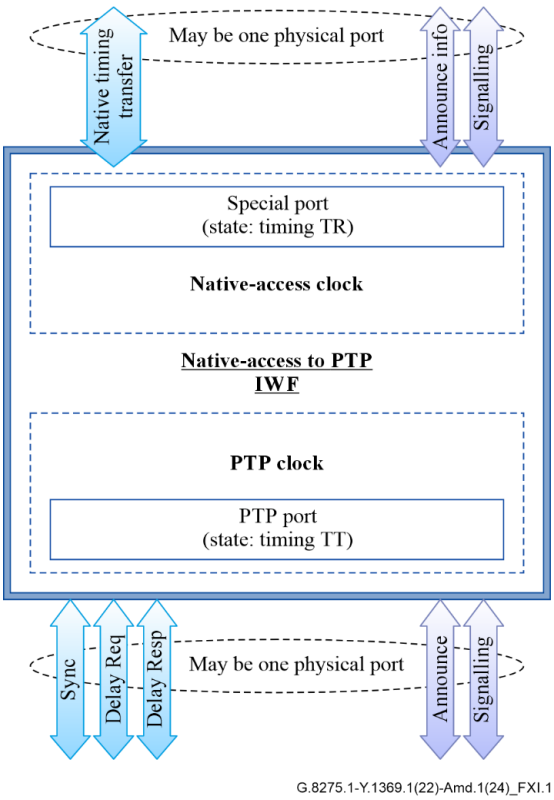


Figure XI.1 – Timing flow between native PTP and native access media

This appendix shows one possible way to model native access media (as a pair of T-BCs). Other ways to model native access media may exist.

Appendix XII

Monitoring alternate timeTransmitter time information provided by a peer PTP port

(This appendix does not form an integral part of this Recommendation.)

The contents of this appendix have been moved into Annex G.

Appendix XIII

Considerations on the use of [IEEE 1588-2019]

(This appendix does not form an integral part of this Recommendation.)

Considerations on the use of [IEEE 1588-2019] are discussed in Appendix IX of [ITU-T G.8275].

Appendix XIV

Considerations in an environment of more than two PTP ports on a single PTP communication path when using transparent clocks and multicast addressing

(This appendix does not form an integral part of this Recommendation.)

There are some aspects of PTP topology that are unique to a multicast environment that uses transparent clocks. When the network is restricted to ordinary and boundary clocks, each transport layer connection has only two PTP ports. On an ordinary clock or boundary clock each local PTP port will only ever communicate with one external PTP port. When transparent clocks are introduced, this situation changes and one local PTP port may communicate with more than one external PTP port.

In Figure XIV.1, PTP clocks A, B, and C are connected using a transparent clock. This makes the three PTP ports A.1, B.1 and C.1 the endpoints of a single PTP communication path. Each of these ports have a portDS dataset within their respective clocks.

In this example, it is assumed that PTP ports A.1 and B.1 either have timeTransmitterOnly set to TRUE or, due to some transport layer distribution of Announce messages (not shown in Figure XIV.1) and/or the rules of the alternate BTCA of [ITU-T G.8275.1], both ports are in the TIME_TRANSMITTER state. Both ports will transmit Announce messages advertising the quality of the grandmaster clocks they are using. Port C.1 will receive the Announce messages from ports A.1 and B.1 due to the transparent clock connection.

According to [IEEE 1588], port C.1 may use a foreign timeTransmitter table and compute an E_{rbest} based on the data of these two flows of Announce messages. In this example, the grandmaster information from port A.1 is better than the grandmaster information from port B.1. The Alternate BTCA in clock C will cause clock A to be the parent PTP instance, then set parentDS.parentPortIdentity to be port A.1 and place port C.1 into the TIME_RECEIVER state.

If there is a problem in the PTP communication between PTP port A.1 and PTP port C.1 then one of the events specified in clause 6.3.9 may be raised by clock C. These PTSF conditions may only apply to that PTP communication; there may be no such condition in the PTP communication between ports B.1 and C.1. Clock C should switch from using port A.1 to port B.1 as its source of timing.

If the PTP communication between Port A.1 and Port C.1 has a PTSF event, the Port C.1 should select the dataset associated with the Port B.1 as its Erbest dataset instead of setting portDS.SF to TRUE. However, if both PTP communications with the Port A.1 and Port B.1 have a PTSF event, the Port C.1 should set portDS.SF to TRUE.

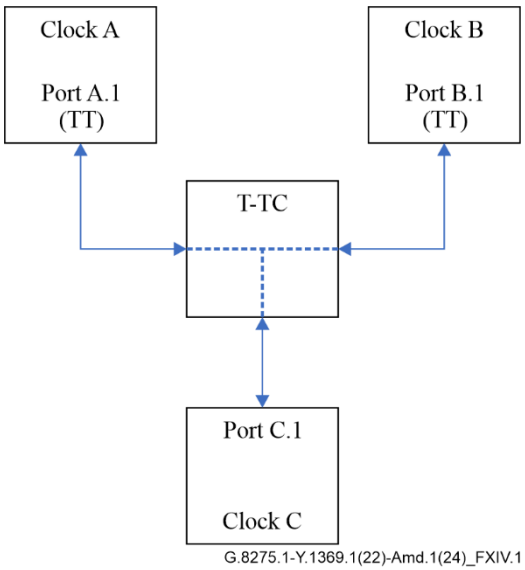


Figure XIV.1 – Three PTP ports interconnected using a transparent clock with multicast addressing

Appendix XV

Deployment considerations for ePRTCs and PRTCs in the network

(This appendix does not form an integral part of this Recommendation.)

Considerations on the deployment of ePRTCs and PRTCs in the network are described in Appendix XV of [ITU-T G.8275].

Appendix XVI

Considerations on the use of clockClass and priority2 in the synchronization network

(This appendix does not form an integral part of this Recommendation.)

In some deployment scenarios, when a T-GM fails, the Alternate BTCA is invoked and it may result in a T-BC in the middle of a chain becoming the new T-GM, and this may not be desirable. This appendix addresses some use-cases and gives some approaches using priority2 and clockClass to achieve the desirable outcome from the Alternate BTCA.

NOTE – Additional considerations on the use of priority2 are provided in Appendix IV of this Recommendation.

Use-Case 1 is depicted in Figure XVI.1. As an example, for this use-case, T-BC 1 does not support holdover within specification, and T-BC 2 to T-BC 20 all support holdover within specification.

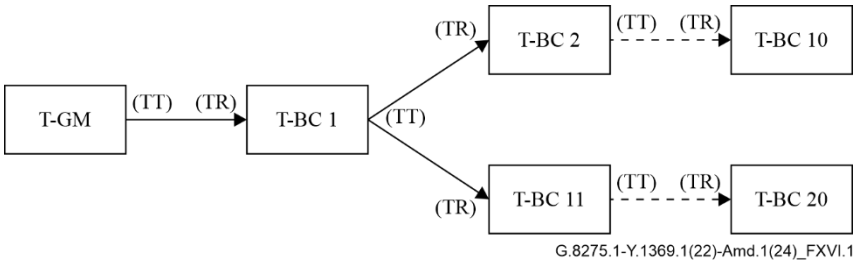


Figure XVI.1 – Use-case 1

If the defaultDS.clockQuality.clockClass of all T-BCs in the locked mode are upgraded to the value of the holdover mode, the clockClass for T-BC 1 would be 165 and for T-BC 2 to T-BC 20 would be 135. When the T-GM fails, T-BC 1 goes into holdover and keeps its clockClass at 165. According to the Alternate BTCA specified in clause 6.3, the T-BC with the lowest clockIdentity within T-BC 2 to T-BC 20 will become the new T-GM, but a desirable topology could be to select T-BC 1 as the new T-GM.

In order to avoid the above issues, one approach is to use defaultDS.priority2 and set the holdover timer of T-BCs (see Note 1 of Table 6) to zero. In Figure XVI.I, the priority2 of T-BC 1 is set to a lower value than the other T-BCs, and the holdover timer of all other T-BCs is set to zero.

Another approach is to set defaultDS.clockQuality.clockClass to 248 for T-BC in the locked mode. If a T-BC loses its input and goes into holdover, according to Appendix VII of this Recommendation, it upgrades defaultDS.clockQuality.clockClass to 135 or 165. After this is done, when the T-GM is lost (see Figure XVI.1), T-BC 1 will be selected as the grandmaster.

Use-Case 2 is depicted in Figure XVI.2. As an example, for this use-case, all T-BCs support holdover within specification.

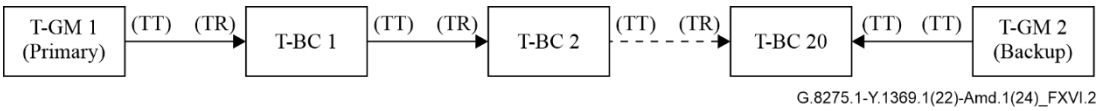


Figure XVI.2 – Use-case 2

Similarly, if the defaultDS.clockQuality.clockClass of all T-BCs in the locked mode are upgraded to the value of the holdover within specification, then the clockClass for all T-BCs would be 135. When T-GM 1 fails, T-BC 1 goes into holdover and keeps its clockClass at 135. According to the Alternate BTCA specified in clause 6.3, the T-BC with the lowest clockIdentity within T-BC 1 to T-BC 20 will become the new T-GM. Before T-GM 2 is selected, based on the order of clockIdentity of all T-BCs, the PTP topology may change several times until the synchronization topology is stable.

In order to avoid the above issues, one approach is to use defaultDS.priority2, and in this case, priority2 of T-BC 1 is set to a lower value than the other T-BCs. Also, to account for failures in the other links, priority2 of each T-BC should be set accordingly to allow the desired synchronization topology.

Another approach is to also set defaultDS.clockQuality.clockClass to 248 for T-BC in the locked mode. If a T-BC loses its input and goes into holdover, according to Appendix VII of this Recommendation, it upgrades defaultDS.clockQuality.clockClass to 135 or 165. After this is done, when the T-GM 1 is lost, then T-BC 1 will become the new T-GM until the T-GM 2 is selected.

Appendix XVII

Aspects of Ethernet switching and T-BC functionality within a network element

(This appendix does not form an integral part of this Recommendation.)

To ensure consistent and correct operation in a PTP network, it is important when configuring a PTP T-BC that all physical ports on the network element carrying PTP messages also have an associated PTP port. This is especially important when using the forwardable PTP MAC address. Not following this guideline can cause instability of the PTP network and/or time errors exceeding the intended performance of the network.

NOTE – If the forwardable address is not in the MAC address filtering database, this is equivalent to having a PTP-unaware device forwarding all ingress PTP messages with the forwardable PTP MAC address, regardless of the value of defaultDS.instanceEnable and portDS.enable.

Bibliography

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